



Theoretical Knowledge

1. Alert Priority Levels

1. Introduction

Alert priority levels are used in a Security Operations Center (SOC) to determine how serious a security alert is and how quickly it needs to be handled. Proper prioritization helps analysts focus on the most critical threats first and ensures efficient incident response.

2. What are Alert Priority Levels?

Alert priority levels classify alerts based on their **severity, impact, and urgency**. Common levels include:

- **Critical** – Immediate action required (e.g., ransomware attack, data breach)
- **High** – Serious threat that can lead to major damage (e.g., unauthorized admin access)
- **Medium** – Suspicious activity that needs investigation (e.g., brute-force login attempts)
- **Low** – Informational or minor issues (e.g., port scan, failed login attempts)

Priority Determination Factors

SOC analysts assign priority based on the following factors:

1. Asset Criticality

- Importance of the affected system
- Example:
 - Production server → High priority
 - Test system → Lower priority

2. Exploit Likelihood

- Chances of attack being successful
- Example:
 - Public exploit available → High risk
 - No known exploit → Lower risk

3. Business Impact

- Effect on organization operations
- Example:
 - Financial loss → High
 - Minor system issue → Low



4. CVSS (Common Vulnerability Scoring System)

CVSS is used to measure the severity of vulnerabilities.

- Score ranges from **0 to 10**
- Higher score = higher risk

Example:

- CVSS **9.8** → Critical
- CVSS **7.5** → High
- CVSS **5.0** → Medium
- CVSS **2.0** → Low

5. Real-World Example

The vulnerability **Log4Shell (CVE-2021-44228)** had a CVSS score of **9.8**, making it a **Critical priority** alert because:

- It allowed remote code execution
- It affected many systems globally
- Exploits were publicly available

6. Tools Used for Alert Prioritization

- **SIEM tools** (e.g., Splunk, Wazuh)
- **Threat intelligence platforms**
- **CVSS scoring systems**

These tools help analysts assign and manage alert priorities efficiently.

7. Conclusion

Alert priority levels are essential for effective SOC operations. By properly classifying and prioritizing alerts, analysts can respond quickly to critical threats, reduce risks, and protect organizational assets.

2. Incident Classification

1. Introduction

Incident Classification is the process of identifying, categorizing, and labeling security incidents based on their type, behavior, and impact. It helps SOC analysts quickly understand threats and take appropriate actions.

2. Objectives

- To classify different types of security incidents
- To standardize incident labeling using frameworks
- To enrich incidents with contextual metadata

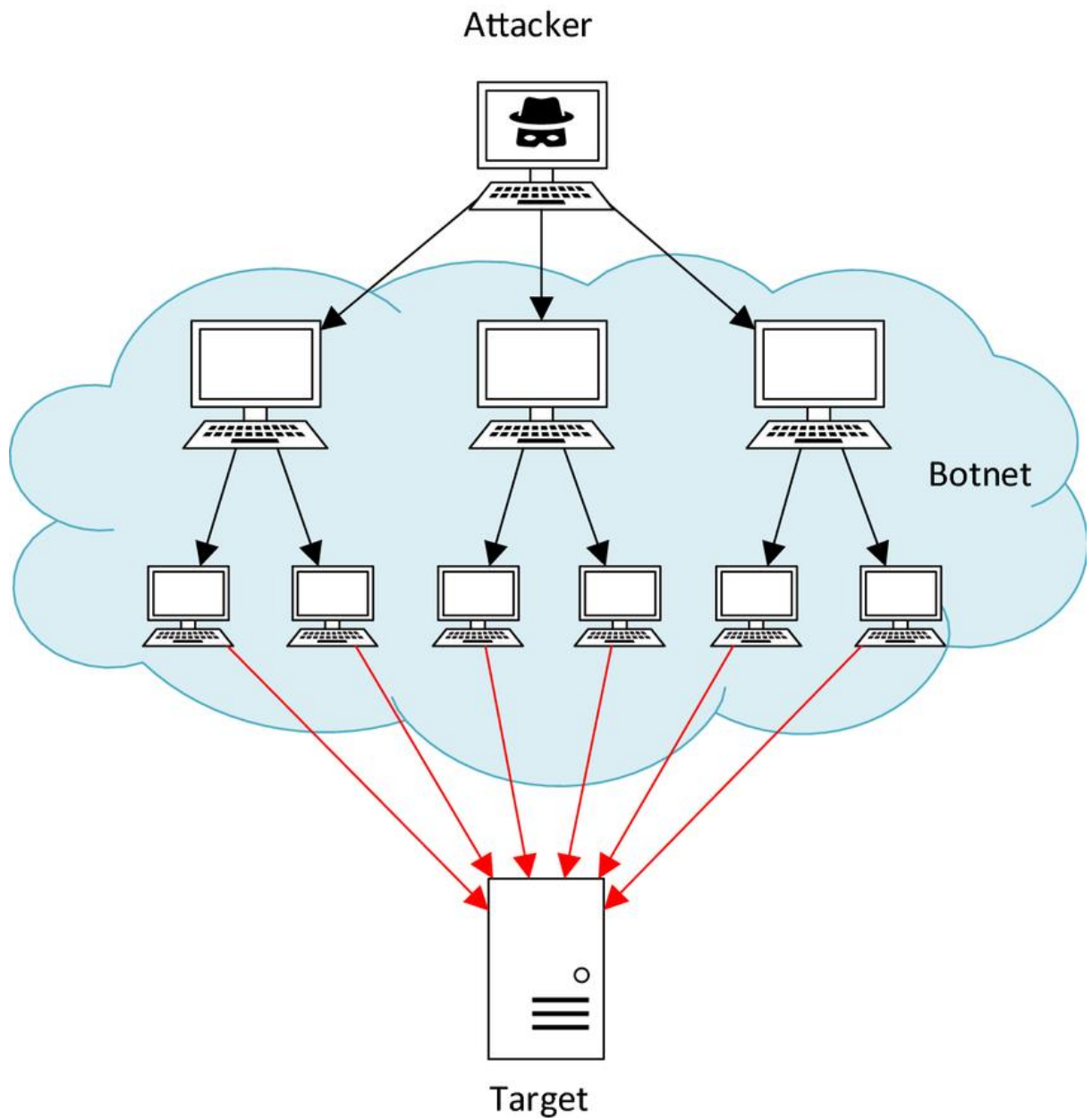


- To improve investigation speed and accuracy

3. Incident Categories

Types of Incidents





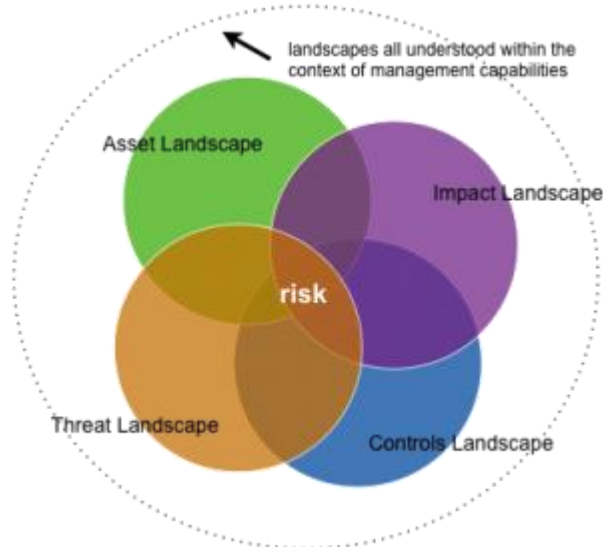
- **Malware** → Malicious software like ransomware, trojans
- **Phishing** → Fake emails/messages to steal credentials
- **DDoS Attack** → Flooding server to make it unavailable
- **Insider Threat** → Employee misuse (e.g., data theft)
- **Data Exfiltration** → Unauthorized data transfer

Example:

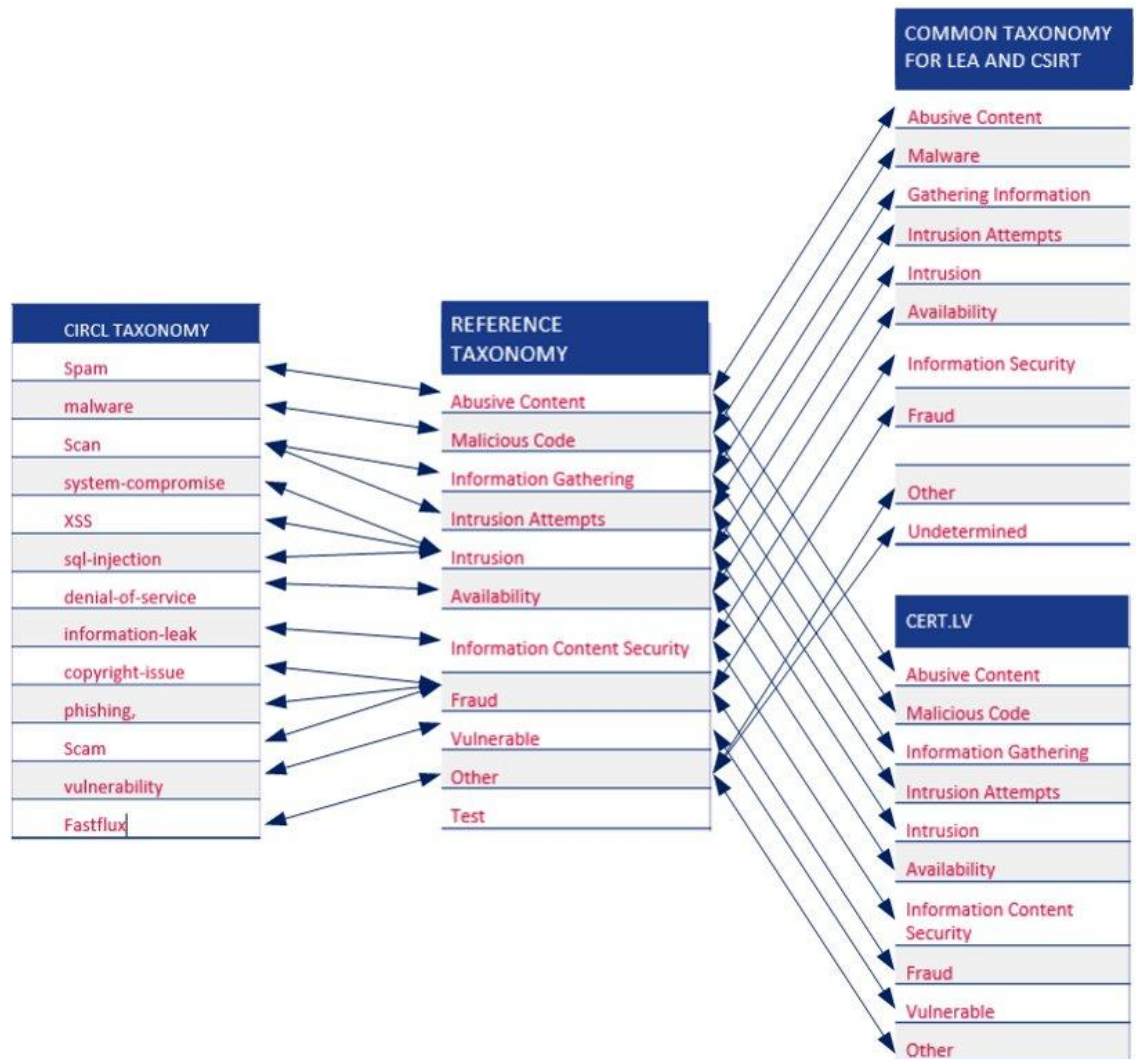
An employee copying company data to USB = **Insider Threat**



4. Taxonomy Frameworks Standard Classification Models



Software Execution x									selection controls		layer controls		technique controls	
Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection						
9 techniques	10 techniques	18 techniques	12 techniques	37 techniques	14 techniques	25 techniques	9 techniques	17 techniques						
Replication Through Removable Media	Native API	BITS Jobs	Process Injection (8/11)	Obfuscated Files or Information (5/5)	Credentials from Password Stores (3/3)	System Information Discovery	Replication Through Removable Media	Screen Capture						
Drive-by Compromise	Windows Management Instrumentation	Hijack Execution Flow (7/11)	Access Token Manipulation (5/5)	Deobfuscate/Decode Files or Information	Network Sniffing	File and Directory Discovery	Data from Local System	Data from Local System						
Valid Accounts (2/4)	Command and Scripting Interpreter (7/8)	Traffic Signaling (0/1)	Exploitation for Privilege Escalation	Modify Registry	OS Credential Dumping (8/8)	Process Discovery	Lateral Tool Transfer	Audio Capture						
Exploit Public-Facing Application	Exploitation for Client Execution	Valid Accounts (2/4)	Hijack Execution Flow (7/11)	Process Injection (8/11)	Brute Force (3/4)	System Network Configuration Discovery	Exploitation of Remote Services	Archive Collected Data (3/3)						
External Remote Services	Shared Modules	Account Manipulation (1/4)	Valid Accounts (2/4)	Rootkit	Steal Web Session Cookie	System Owner/User Discovery	Taint Shared Content	Clipboard Data						
Hardware Additions	Scheduled Task/Job (3/6)	Browser Extensions	Boot or Logon Autostart Execution (8/12)	Indicator Removal on Host (5/6)	Two-Factor Authentication Interception	Query Registry	Remote Services (6/6)	Video Capture						
Phishing (2/3)	Software Deployment Tools	Boot or Logon Initialization Scripts (3/5)	Group Policy Modification	Access Token Manipulation (5/5)	Unsecured Credentials (4/6)	System Network Connections Discovery	Automated Collection	Automated Collection						
Supply Chain Compromise (1/3)	Inter-Process Communication (2/2)	Compromise Client Software Binary	Scheduled Task/Job (3/6)	Virtualization/Sandbox Evasion (3/3)	Exploitation for Credential Access	System Time Discovery	Data from Removable Media	Data from Removable Media						
Trusted Relationship	System Services (2/2)	External Remote Services	Abuse Elevation Control Mechanism (4/4)	BITS Jobs	Forced Authentication	System Service Discovery	Man in the Browser	Man in the Browser						
	User Execution (2/2)	Scheduled Task/Job (3/6)	Boot or Logon Initialization Scripts (3/5)	Hijack Execution Flow (7/11)	Input Capture (3/4)	Peripheral Device Discovery	Data from Network Shared Drive	Data from Network Shared Drive						
		Boot or Logon Initialization Scripts (3/5)	Create or Modify System Process (4/4)	Masquerading (5/6)	Man-in-the-Middle (1/2)	Remote System Discovery	Data from Cloud Storage Object	Data from Cloud Storage Object						
		Create Account (2/3)	Event Triggered Execution (10/15)	Traffic Signaling (0/1)	Modify Authentication Process (3/4)	Application Window Discovery	Use Alternate Authentication Material (2/4)	Use Alternate Authentication Material (2/4)						
		Create or Modify System Process (4/4)		Valid Accounts (2/4)	Steal Application Access Token	Network Service Scanning								
		Event Triggered Execution (10/15)		Indirect Command Execution	Steal or Forge Kerberos Tickets (3/4)	Network Share Discovery								
		Implant Container Image		XSL Script Processing		Software Discovery (1/1)								
				Abuse Elevation Control Mechanism (4/4)		Network Sniffing								
				Direct Volume Access		Domain Trust								



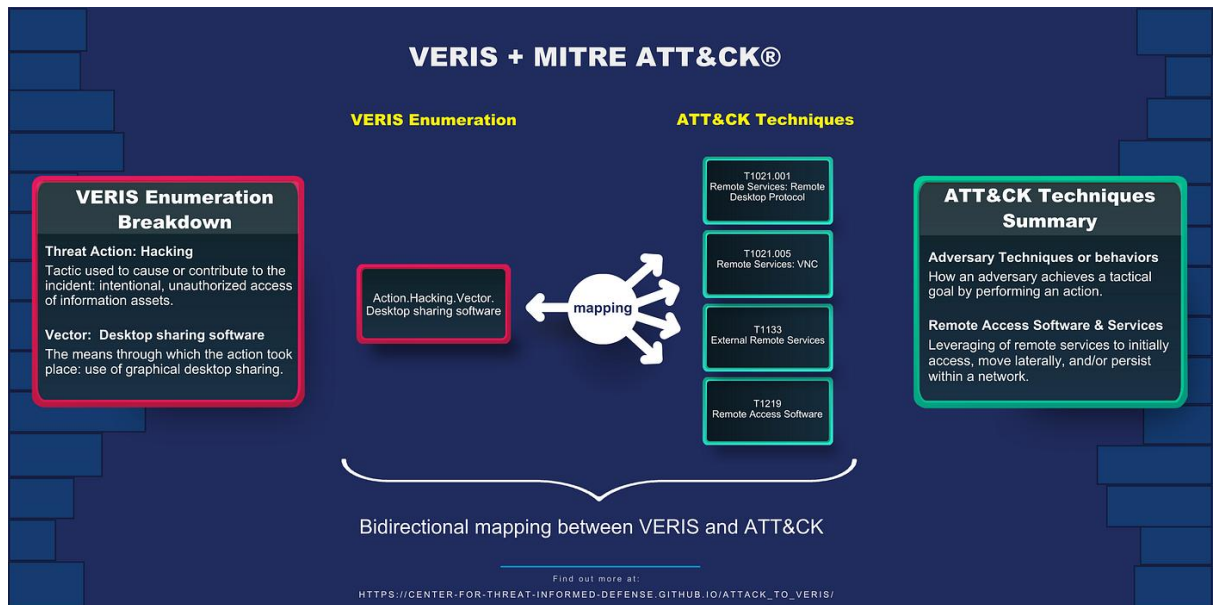


VERIS Model

Bringing Structure to Cybersecurity



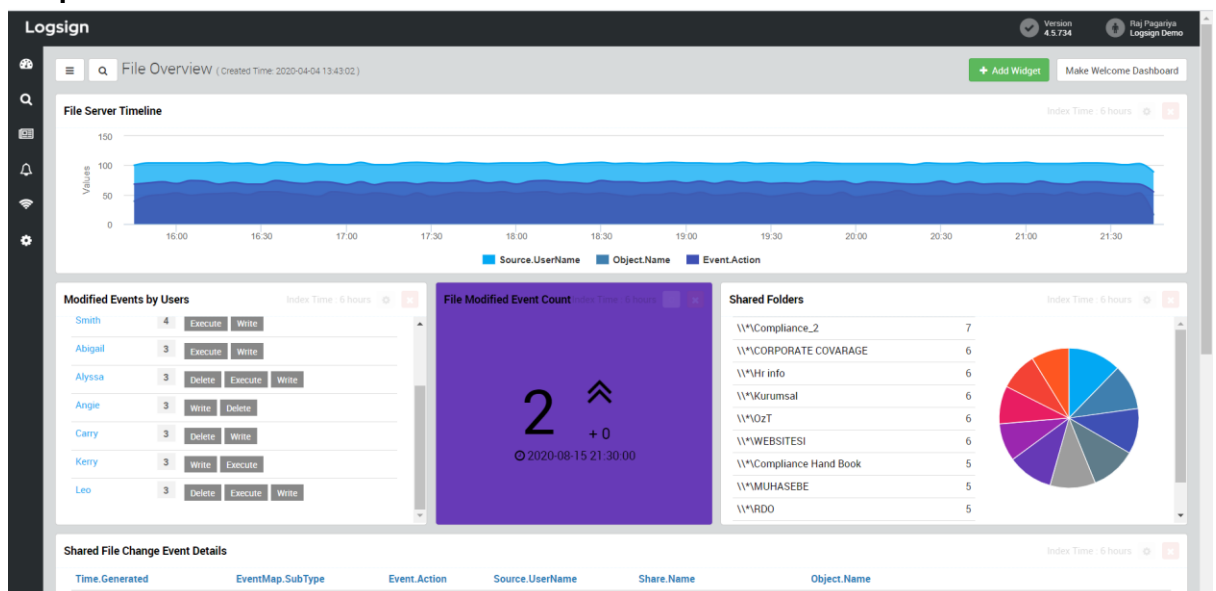
From fragmented data to actionable intelligence



- **MITRE ATT&CK**
 - Maps attacker behavior
 - Example: **T1566** → **Phishing**
- **ENISA Incident Taxonomy**
 - Standard European classification model
 - Categorizes incidents by type and impact
- **VERIS Framework**
 - Used for structured incident reporting
 - Focus on threat actor, action, asset, impact

5. Contextual Metadata

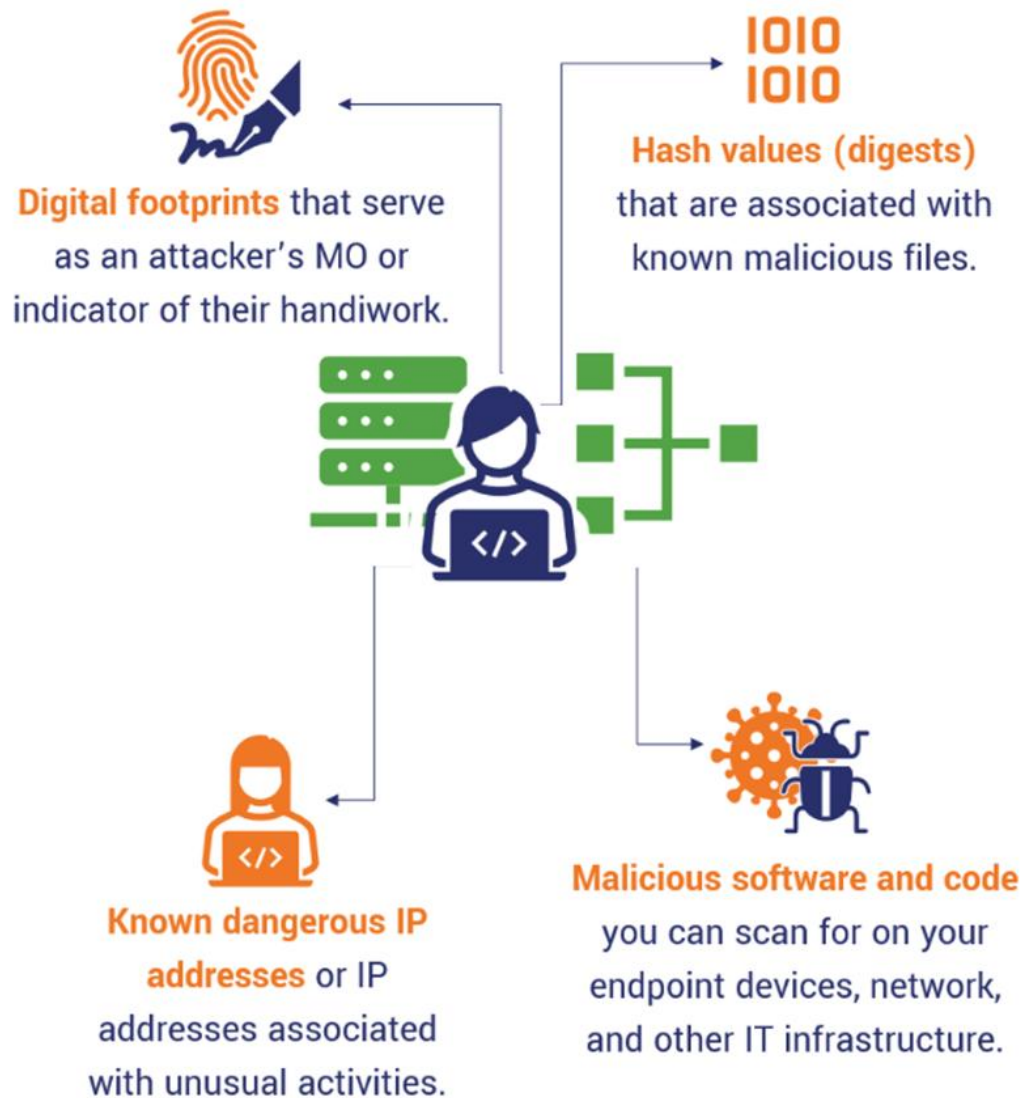
Important Metadata Fields

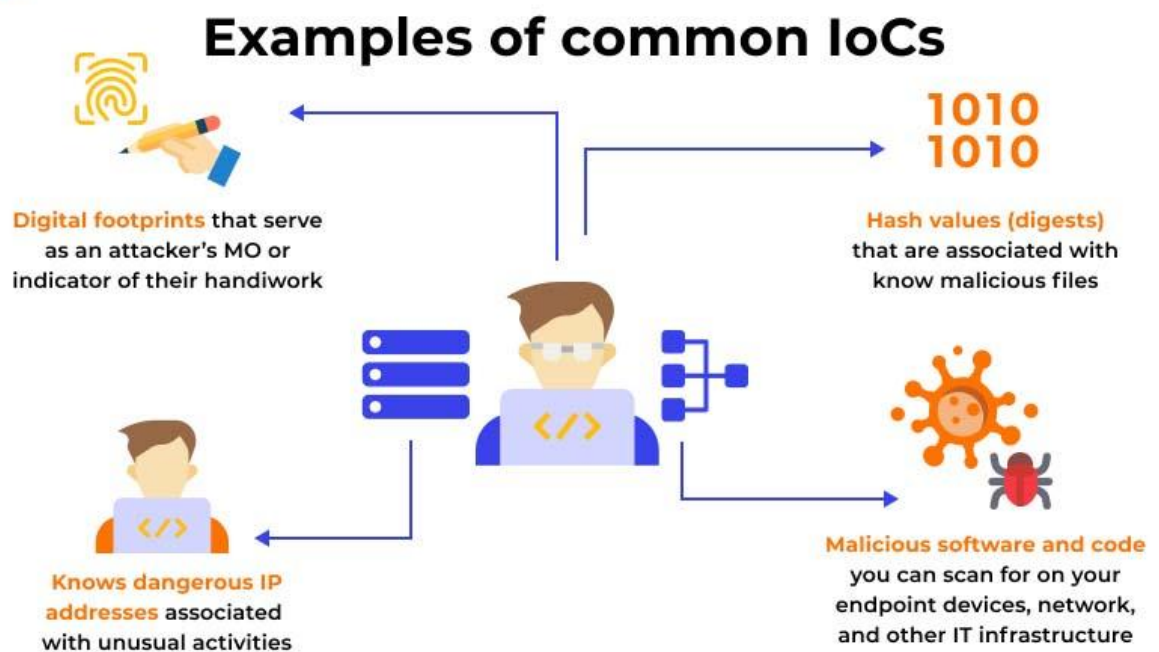
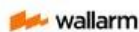
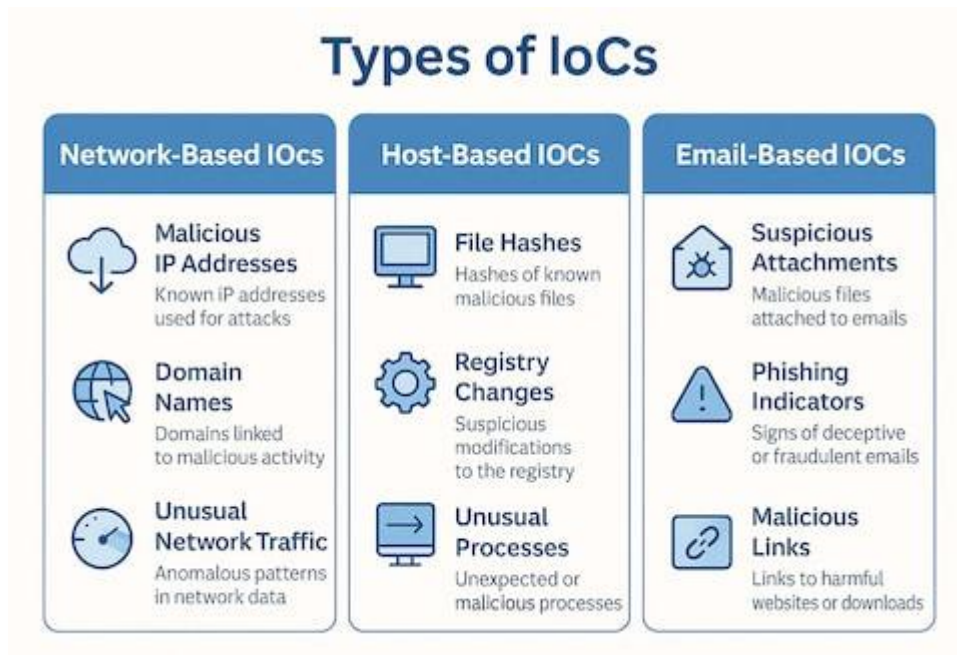






Indicators of Compromise





- Timestamp → When incident occurred
- Source/Destination IP
- Affected systems
- User details
- IOCs (Indicators of Compromise)
 - Malicious hash
 - Suspicious domain
 - IP address

👉 Example:



- IP: 192.168.1.10
- Hash: abc123xyz
- Time: 10:30 AM

6. Incident Classification Process

1. Identify the alert
2. Analyze logs and evidence
3. Categorize the incident (e.g., phishing, malware)
4. Map to framework (MITRE ATT&CK)
5. Add metadata (IP, timestamp, IOCs)
6. Assign severity level

7. Case Study Example

Scenario:

User receives fake email asking for login credentials

Analysis:

- Type → Phishing
- MITRE → T1566
- Metadata → Email sender, IP, timestamp

Conclusion:

Incident classified as **Phishing Attack**

8. Conclusion

Incident classification is a critical step in SOC operations. By using frameworks like MITRE ATT&CK and enriching incidents with metadata, analysts can quickly detect, analyze, and respond to threats effectively.

TASK 3 : Basic Incident Response

Introduction

Incident Response (IR) is a structured approach used by Security Operations Center (SOC) teams to detect, analyze, and respond to cybersecurity incidents. It helps organizations minimize damage, recover quickly, and prevent future attacks. Industry frameworks such as NIST SP 800-61 and SANS Institute guidelines are widely used for effective incident handling.

Incident Response Lifecycle

Preparation → Identification → Containment → Eradication → Recovery → Lessons Learned



1 Preparation

- > Creation of playbooks
- > Deployment of tools (SIEM, EDR)
- > Team training

A playbook is a step-by-step guide used to handle specific incidents like phishing or malware.

2 Identification

- Alert monitoring using SIEM
- Alert triage (High, Medium, Low)
- False positive filtering

Example:

Multiple failed login attempts detected → Investigated for brute-force attack.

3 Containment

- Isolate infected systems
- Block malicious IPs
- Disable compromised accounts

Endpoint Detection and Response (EDR) tools are used for isolation.

4 Eradication

- Remove malware
- Delete malicious files
- Patch vulnerabilities

Example:

Trojan malware removed using antivirus and manual cleanup

5 Recovery

- Restore system from backup
- Monitor system
- Reconnect to network

6 Lessons Learned

Post-incident analysis is conducted to identify the root cause and improve future security measures.

Incident Response Procedures

1 System Isolation



Infected System → Disconnect Network → Block IP → Disable User → Secure System

- Disconnect network
- Use EDR tools
- Prevent lateral movement

2 Evidence Preservation

- Memory dump collection
- Log analysis
- File hashing (SHA256)

3 Communication Protocols

- SOC Analyst → SOC Lead
- SOC Lead → Incident Response Team
- Incident reporting & documentation

4 SOAR Tools

Alert → SOAR Tool → Auto Analysis → Action (Block/Isolate) → Report

SOAR tools like Splunk Phantom automate incident response tasks such as alert triage and threat containment.

- **NIST SP 800-61:** Provides standard incident response lifecycle and framework
- **SANS Handbook:** Gives practical templates and real-world response techniques
- **LetsDefend:** Offers hands-on SOC simulation and SOAR learning

Tools & Frameworks Used

- SIEM tools (Splunk, Wazuh)
- EDR tools (Defender, CrowdStrike)
- SOAR tools
- Frameworks:
 - NIST SP 800-61
 - SANS Incident Handler Handbook

Conclusion

This task provided practical understanding of how SOC teams respond to cybersecurity incidents. It enhanced my knowledge of incident handling procedures, tools, and industry standards, preparing me for real-world SOC operations.



Practical Application

1. Alert Management Practice

Introduction

Alert management is a critical function in a Security Operations Center (SOC), where security alerts are analyzed, prioritized, and responded to efficiently. This task focuses on understanding alert classification, prioritization using CVSS, mapping with MITRE ATT&CK, and handling incidents using tools like Wazuh and TheHive.

Alert Classification System

An alert classification system was created using Google Sheets to categorize alerts based on type, severity, and MITRE ATT&CK techniques.

The table includes:

- Alert ID
- Alert Name
- Type
- CVSS Score
- Priority
- MITRE Technique

Alert ID	Alert Name	Type	Real Event ID / Rule	CVSS	Priority	MITRE
1	Phishing Email	Phishing	Email Gateway Alert / Rule ID 100001	8	High	T1566
2	Log4Shell Exploit	Vulnerability Exploit	Wazuh Rule ID 31168 (Log4j exploit)	9.8	Critical	T1190
3	Port Scan Detected	Reconnaissance	Wazuh Rule ID 5710	3.5	Low	T1046
4	Failed Login Attempt	Authentication	Windows Event ID 4625	5	Medium	T1110
5	Successful Login	Authentication	Windows Event ID 4624	3	Low	T1078
6	Brute Force Attack	Credential Attack	Multiple 4625 events / Wazuh Rule 5715	7.5	High	T1110
7	Malware Execution	Execution	Windows Event ID 4688 (process creation)	8.5	High	T1204
8	Account Lockout	Account Security	Windows Event ID 4740	6.5	Medium	T1110
9	Privilege Escalation	Privilege Abuse	Windows Event ID 4672	9	Critical	T1068
10	PowerShell Suspicious Activity	Execution	Windows Event ID 4104	8	High	T1059

Figure 1 : Alert Classification Table



Alert Prioritization

Alerts were prioritized using CVSS scoring:

- Critical → 9–10
- High → 7–8
- Medium → 4–6
- Low → 0–3

Example:

- Log4Shell → CVSS 9.8 → Critical
- Port Scan → CVSS 3.5 → Low

MITRE ATT&CK Mapping

Alerts were mapped to MITRE ATT&CK techniques:

- Phishing → T1566
- Exploit → T1190
- Brute Force → T1110
- Port Scan → T1046

This helps identify attacker behavior and tactics.

Wazuh Dashboard & Log Monitoring

Wazuh SIEM was used to monitor real-time logs and security events. The dashboard provides visibility into:

- Alert counts
- Event logs
- Rule IDs
- Severity levels

The Events tab shows logs such as:

- Port status changes
- Login sessions
- AppArmor denied events

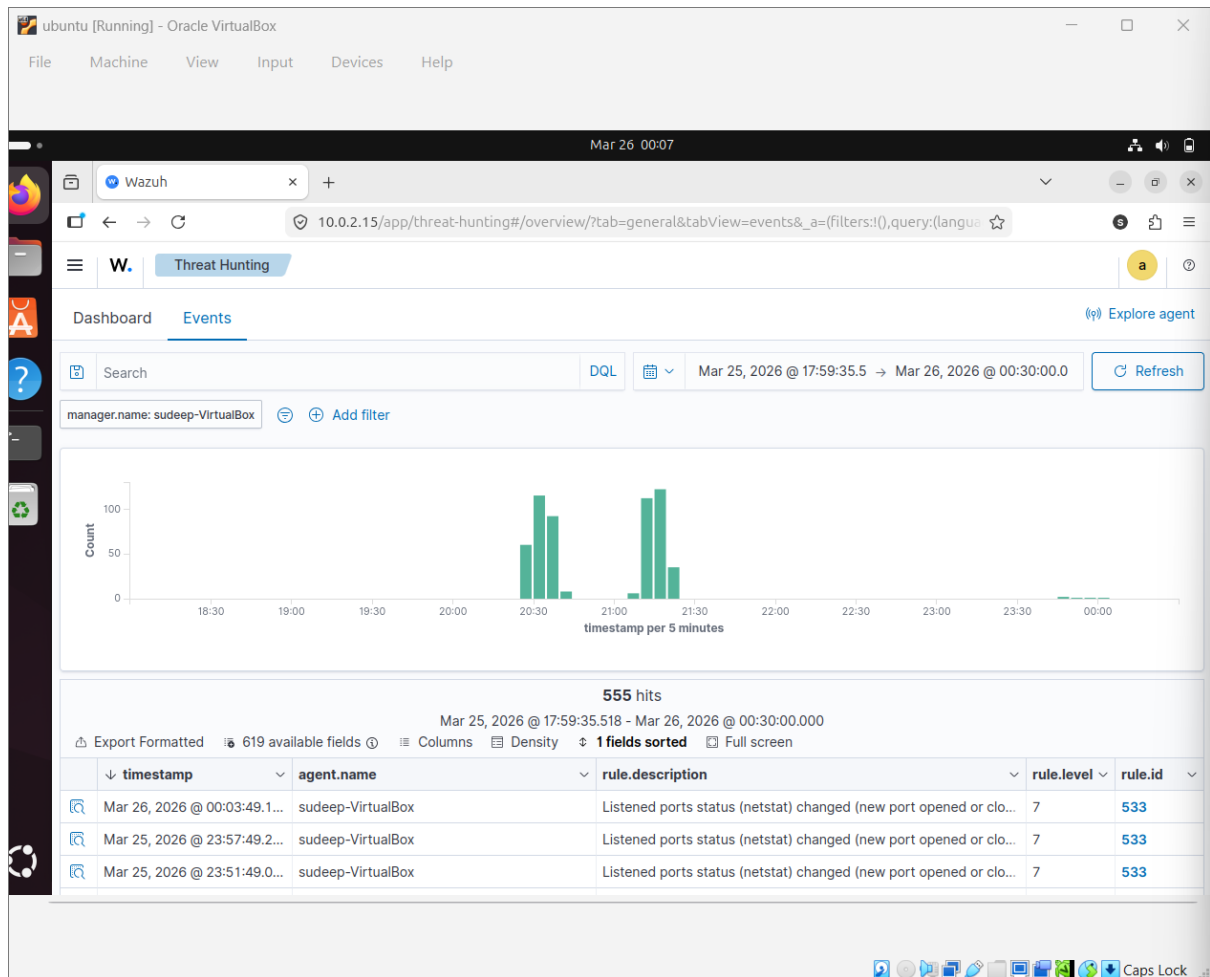


Figure 2 : Wazuh Logs Table



ubuntu [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Mar 26 00:08

Wazuh

10.0.2.15/app/threat-hunting#/overview/?tab=general&tabView=events&a=(filters:(),query:(language

Threat Hunting

Mar 25, 2026 @ 17:59:35.518 - Mar 26, 2026 @ 00:30:00.000

Export Formatted 619 available fields Columns Density 1 fields sorted Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Mar 26, 2026 @ 00:03:49.1...	sudeep-VirtualBox	Listened ports status (netstat) changed (new port opened or clo...	7	533
Mar 25, 2026 @ 23:57:49.2...	sudeep-VirtualBox	Listened ports status (netstat) changed (new port opened or clo...	7	533
Mar 25, 2026 @ 23:51:49.0...	sudeep-VirtualBox	Listened ports status (netstat) changed (new port opened or clo...	7	533
Mar 25, 2026 @ 23:47:02.6...	sudeep-VirtualBox	PAM: Login session opened.	3	5501
Mar 25, 2026 @ 23:46:01.3...	sudeep-VirtualBox	Wazuh server started.	3	502
Mar 25, 2026 @ 21:21:39.0...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:35.0...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:28.9...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:22.9...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:18.9...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:12.9...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:12.9...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:08.9...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:08.9...	sudeep-VirtualBox	Apparmor DENIED	3	52002
Mar 25, 2026 @ 21:21:02.8...	sudeep-VirtualBox	Apparmor DENIED	3	52002

Rows per page: 15

< 1 2 3 4 5 ... 37 >

Figure 3 : Wazuh Events Page

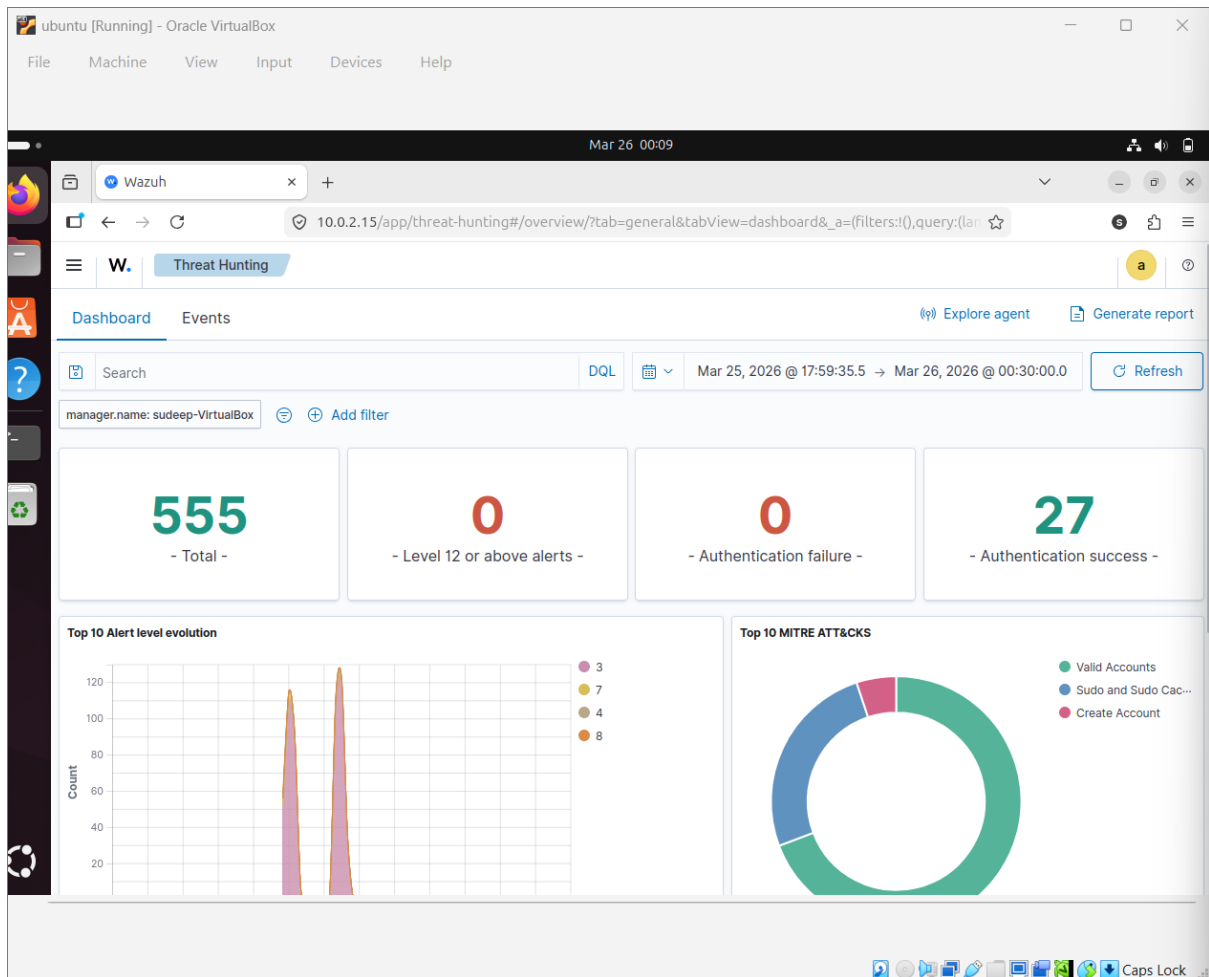


Figure 4 Dashboard Overview

Dashboard Visualization

A custom dashboard was created in Wazuh to visualize alert severity.

- Pie chart → Alert severity distribution
- Timeline graph → Event trends

This helps in quick analysis and decision-making.

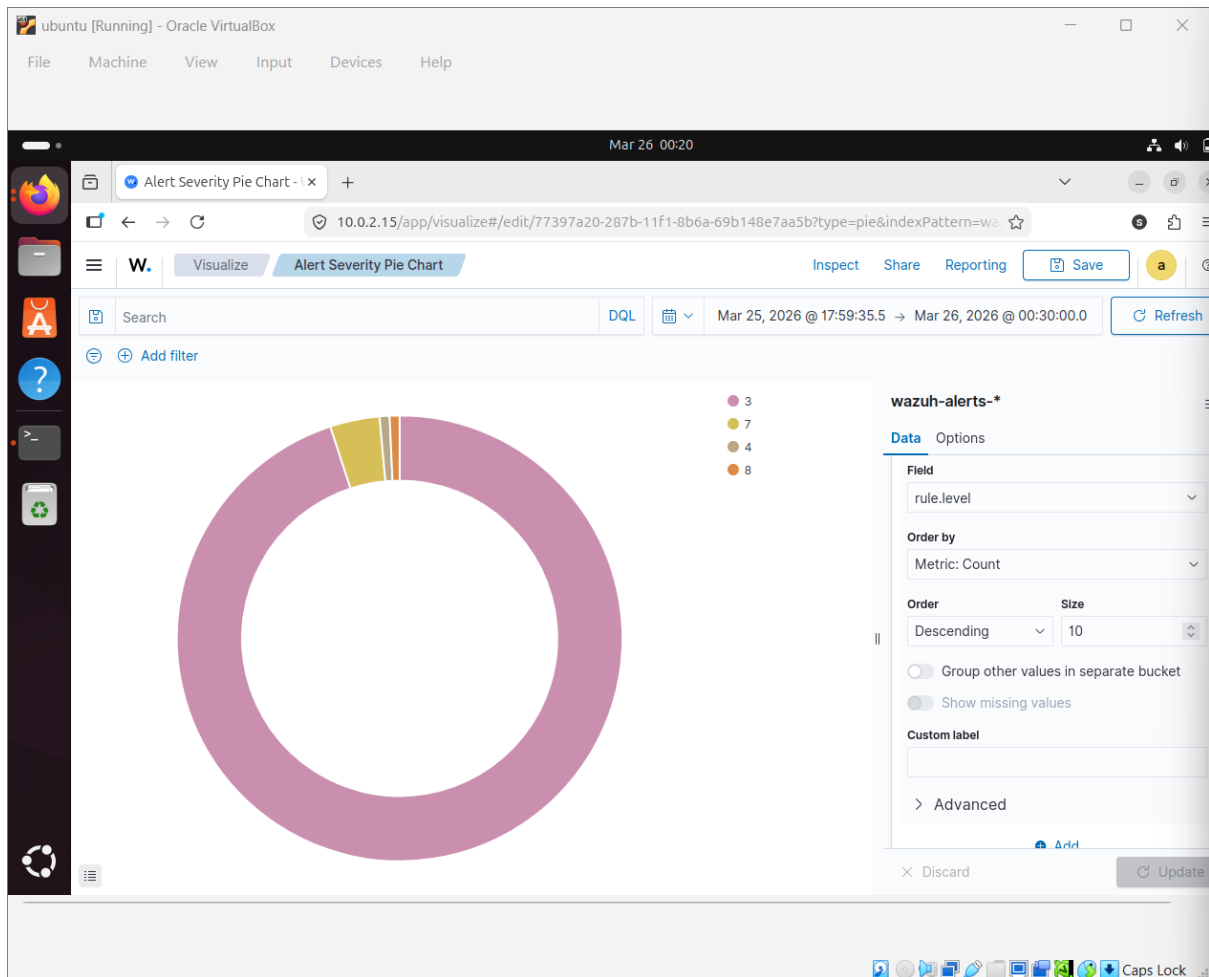


Figure 5 Alert Severity Pie Chart Dashboard

Incident Ticket Creation (TheHive)

Case Details:

- Title: [Critical] Ransomware Detected on Server-X
- Severity: Critical
- Assignee: SOC Analyst

Indicators:

- File: crypto_locker.exe
- IP Address: 192.168.1.50

Observables and tasks were added for investigation.

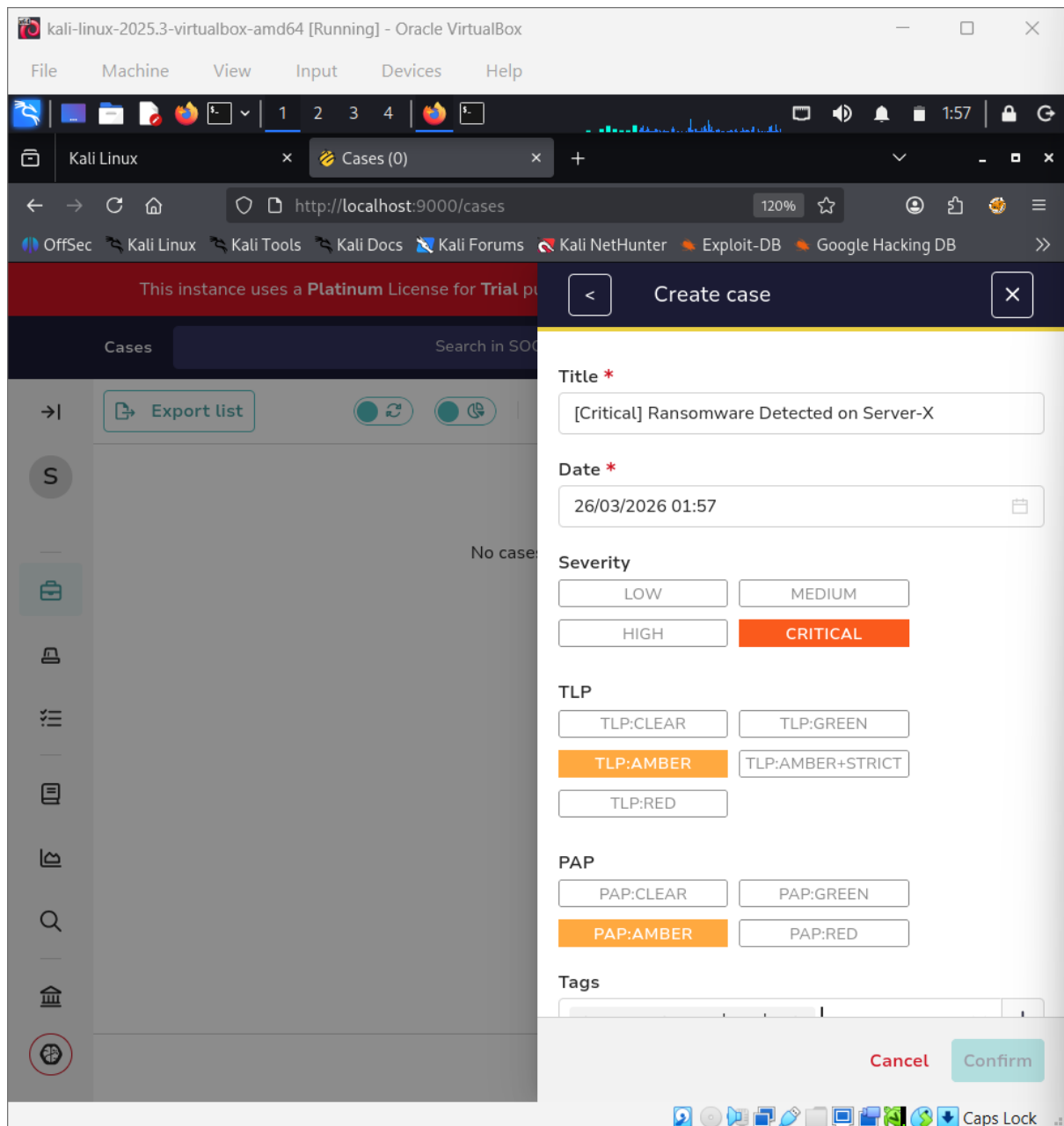


Figure 6 Case Creation

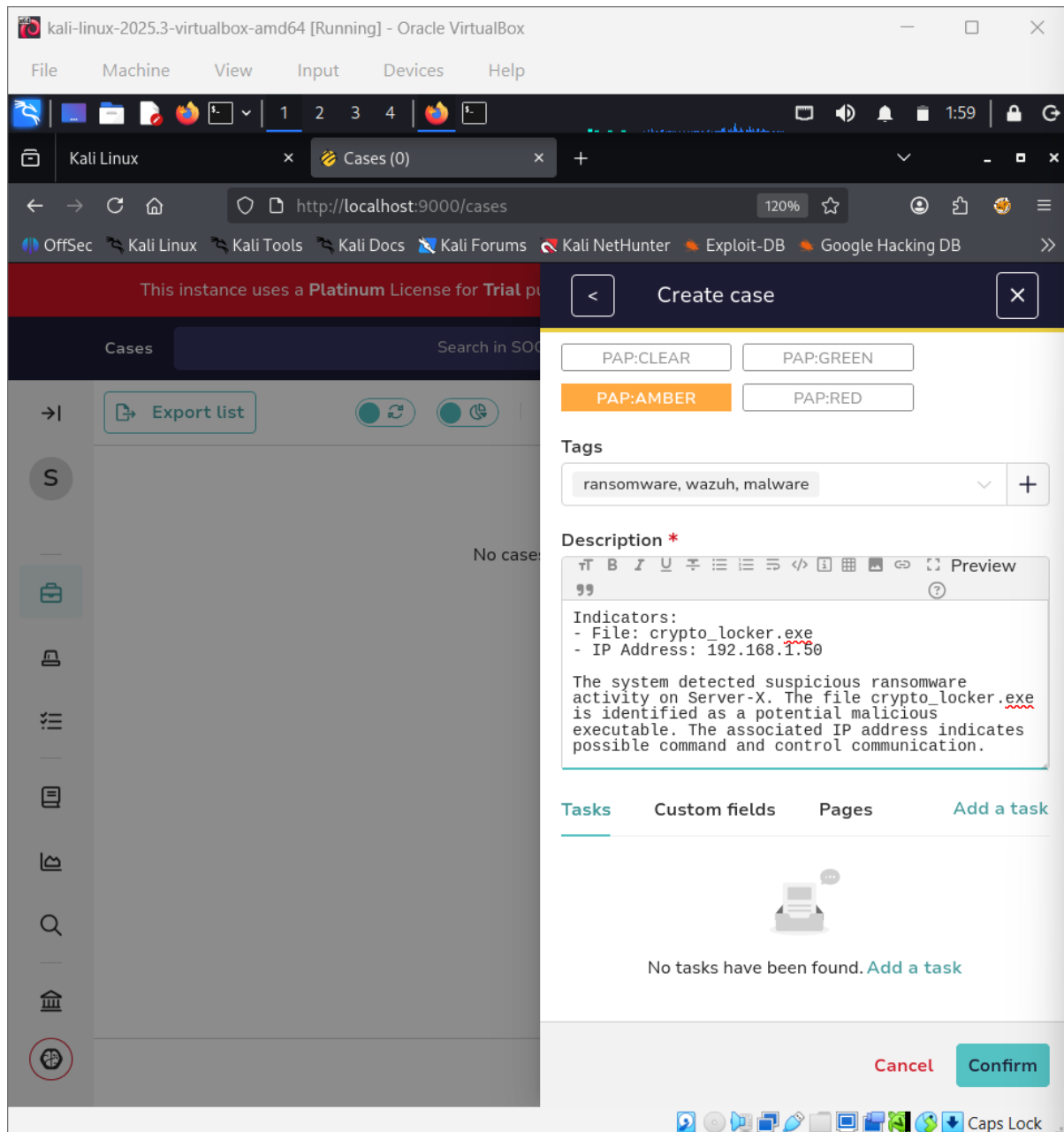


Figure 7 Description



The screenshot shows a web browser window displaying the CYART interface. The browser address bar shows `http://localhost:9000/cases/~24664/observables`. The interface has a red banner at the top stating: "This instance uses a **Platinum License** for **Trial** purpose, and will expire in **15 days**. [Register now.](#)"

The main content area shows a case titled "#1 [Critical] Ransomware Detected on Server-X". The case details on the left include:

- id: ~24664
- Created by: task1@task1.co
- Created at: 26/03/2026 02:00
- Assignee: SOC Analyst
- Status: New
- Start date: 26/03/2026 01:57
- Tasks completion: (empty)

The observables table on the right has the following columns: Flags, Data type, Value/Filename, Dates, C., U., and a menu icon. The table contains one row:

Flags	Data type	Value/Filename	Dates	C.	U.	
☐ TLP:AMBER ☐ PAP:AMBER	filename	crypto_locker[.]exe	S. 26/03/2026 02:08 C. 26/03/2026 02:08			...

Below the table, it says "None" and "No report(s) available". The bottom of the interface shows pagination: "< Previous 0 - 1 of 1 Next > Show 30".

Figure 7 : Observables filename

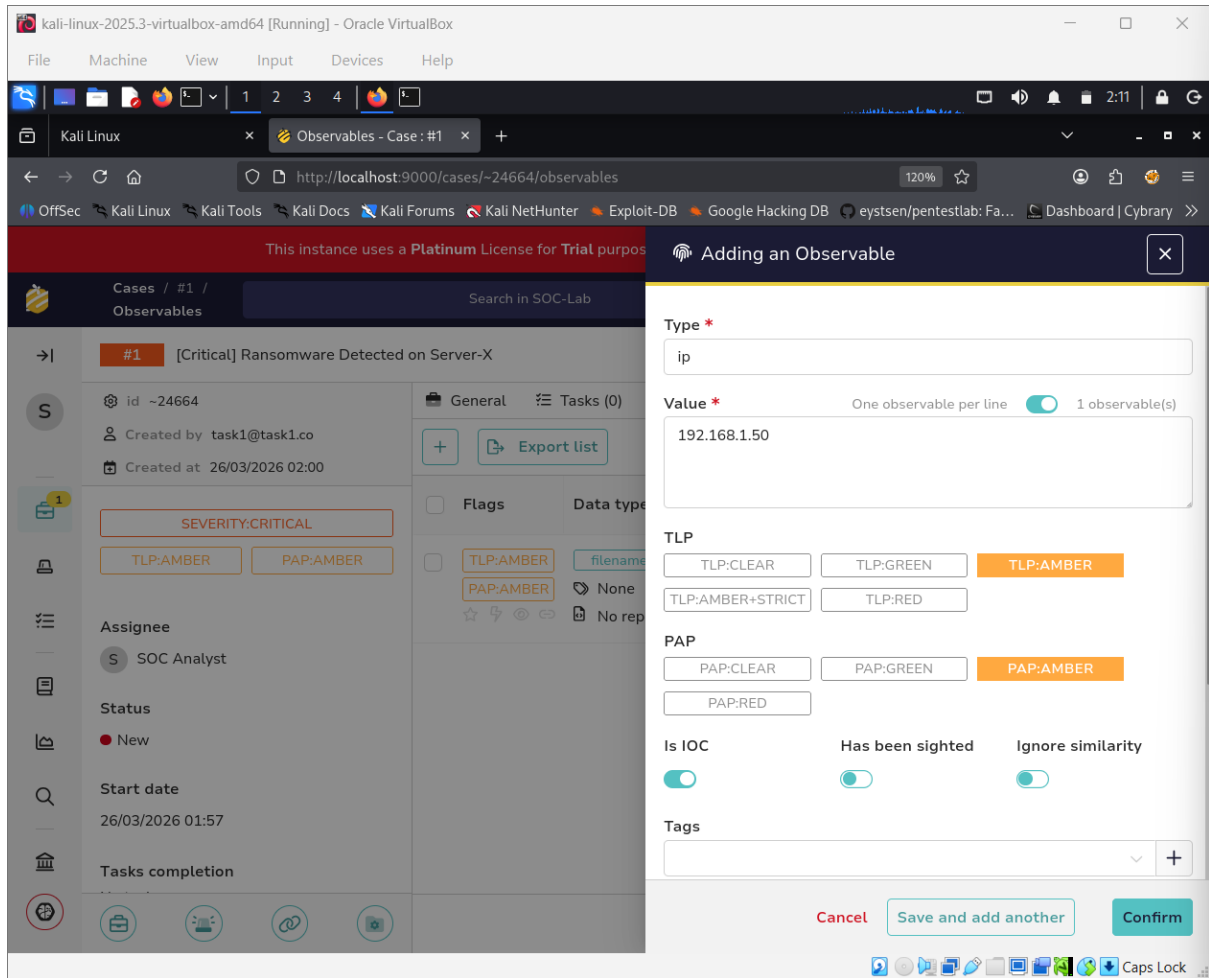


Figure 8 Observables IP

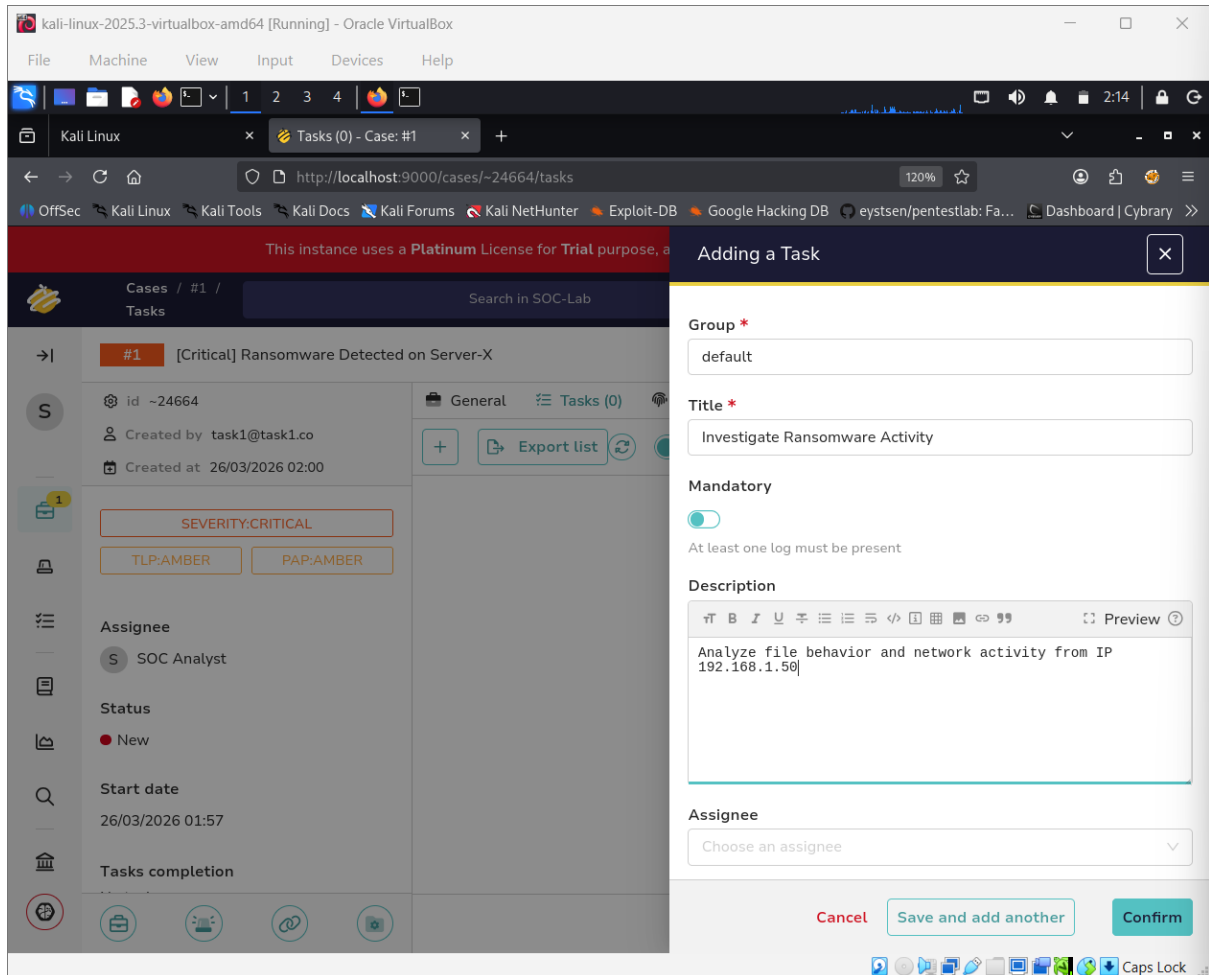


Figure 9 Task Creation

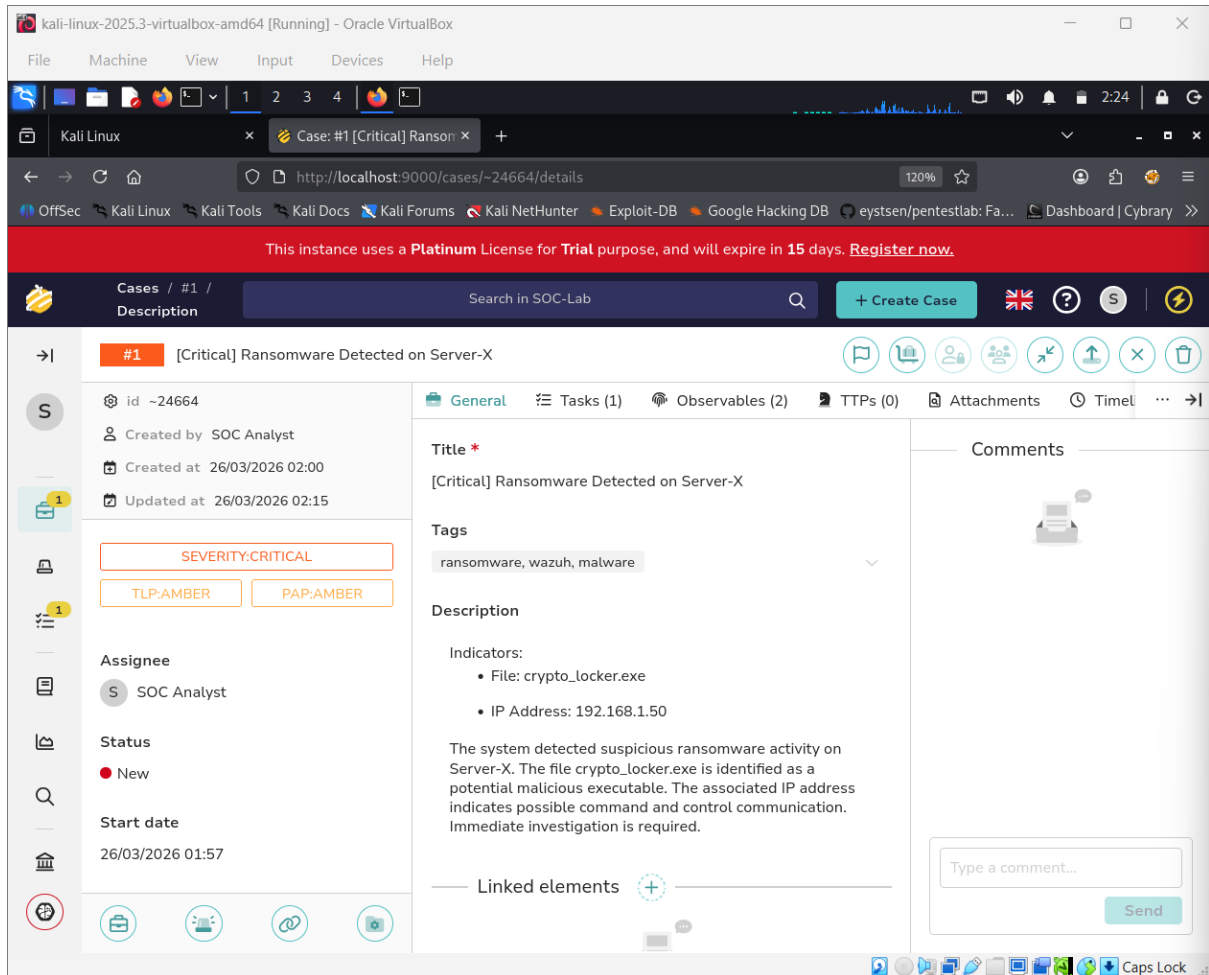


Figure 10 Final Case View



Escalation Process

A formal escalation email was drafted to notify Tier 2 analysts about the critical incident.

Dear Tier 2 Team,

A critical ransomware alert has been detected on Server-X through Wazuh monitoring. The alert indicates suspicious activity involving a malicious file named "crypto_locker.exe" and communication from IP address 192.168.1.50.

These indicators suggest potential ransomware execution and possible command-and-control activity. Immediate investigation is required to prevent further impact.

Kindly analyze the file, block the IP if malicious, and provide updates on mitigation actions.

Regards,

SOC Analyst

Conclusion

This task provided hands-on experience in alert classification, prioritization using CVSS, MITRE ATT&CK mapping, real-time log monitoring using Wazuh, incident ticket creation in TheHive, and escalation procedures. It helped in understanding the real-world SOC workflow and incident response process.



3.Alert Triage Practice

Tools Used

- Wazuh SIEM
- Linux (Ubuntu VM)
- VirusTotal
- GuardPot (Threat Intelligence)

Practical Execution

- Used:

`sudo tail -f /var/log/auth.log`

- Observed:
 - Authentication logs
 - System login events
 - Failed login attempts



The screenshot shows a terminal window titled 'ubuntu [Running] - Oracle VirtualBox'. Inside the terminal, a user named 'sudeep' has run the command 'sudo tail -f /var/log/auth.log'. The output shows a series of authentication events, including a session being logged out, a session being removed, and a session being opened for the user 'root' by 'sudeep'. The logs are timestamped and include details about the user, the command, and the session ID.

```
sudeep@sudeep-VirtualBox:~$ sudo tail -f /var/log/auth.log
[sudo] password for sudeep:
2026-03-26T14:29:25.667539+05:30 sudeep-VirtualBox polkitd[692]: Unregistered Authentication Agent for unix-session:c1 (system bus name :1.45, object path /org/freedesktop/PolicyKit1/AuthenticationAgent, locale en_US.UTF-8) (disconnected from bus)
2026-03-26T14:29:25.721427+05:30 sudeep-VirtualBox systemd-logind[704]: Session c1 logged out. Waiting for processes to exit.
2026-03-26T14:29:25.731402+05:30 sudeep-VirtualBox systemd-logind[704]: Removed session c1.
2026-03-26T14:29:35.928011+05:30 sudeep-VirtualBox (sd-pam): pam_unix(systemd-user:session): session closed for user gdm
2026-03-26T14:30:02.069553+05:30 sudeep-VirtualBox CRON[3508]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-03-26T14:30:02.077591+05:30 sudeep-VirtualBox CRON[3508]: pam_unix(cron:session): session closed for user root
2026-03-26T14:30:27.207183+05:30 sudeep-VirtualBox pkexec: pam_unix(polkit-1:session): session opened for user root(uid=0) by sudeep(uid=1000)
2026-03-26T14:30:27.211794+05:30 sudeep-VirtualBox pkexec[3558]: sudeep: Executing command [USER=root] [TTY=unknown] [CWD=/home/sudeep] [COMMAND=/usr/lib/update-notifier/package-system-locked]
2026-03-26T14:30:59.389245+05:30 sudeep-VirtualBox sudo: sudeep : TTY=pts/0 ; PWD=/home/sudeep ; USER=root ; COMMAND=/usr/bin/tail -f /var/log/auth.log
2026-03-26T14:30:59.397351+05:30 sudeep-VirtualBox sudo: pam_unix(sudo:session): session opened for user root(uid=0) by sudeep(uid=1000)
```

Figure 1 Linux Authentication Logs Monitoring

Simulated Attack (SSH Failed Login)

- Performed:
ssh fakeuser@localhost
- Multiple incorrect password attempts generated logs

Alert Analysis in Wazuh

Observed alerts:

- **AppArmor DENIED (Low priority)**
- **SSH authentication failure (High priority)**



Key Fields:

- rule.id
- rule.level
- rule.description
- srcip (127.0.0.1)

Alert Triage Table

Alert ID	Description	Source IP	Rule ID	Rule Level	Priority	Status
001	AppArmor DENIED	Localhost	52002	3	Low	Closed
002	SSH Authentication Failure	127.0.0.1	2502	10	High	

Analysis of Alerts

♦ Low Priority Alert

- AppArmor DENIED
- Normal system restriction
- No malicious activity
- Classified as **False Positive / Informational**

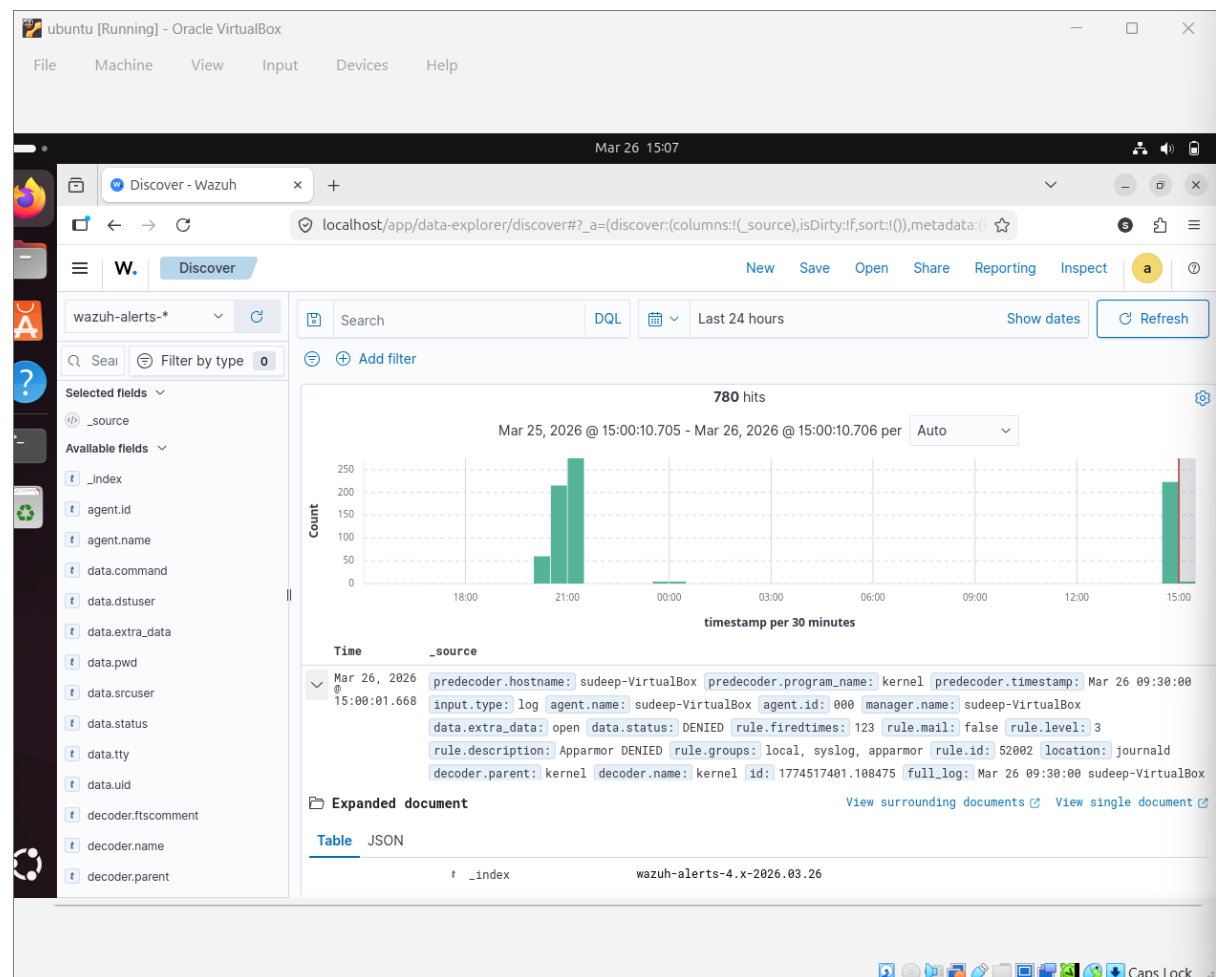


Figure 2 Wazuh Alert – Low Severity (AppArmor)-a

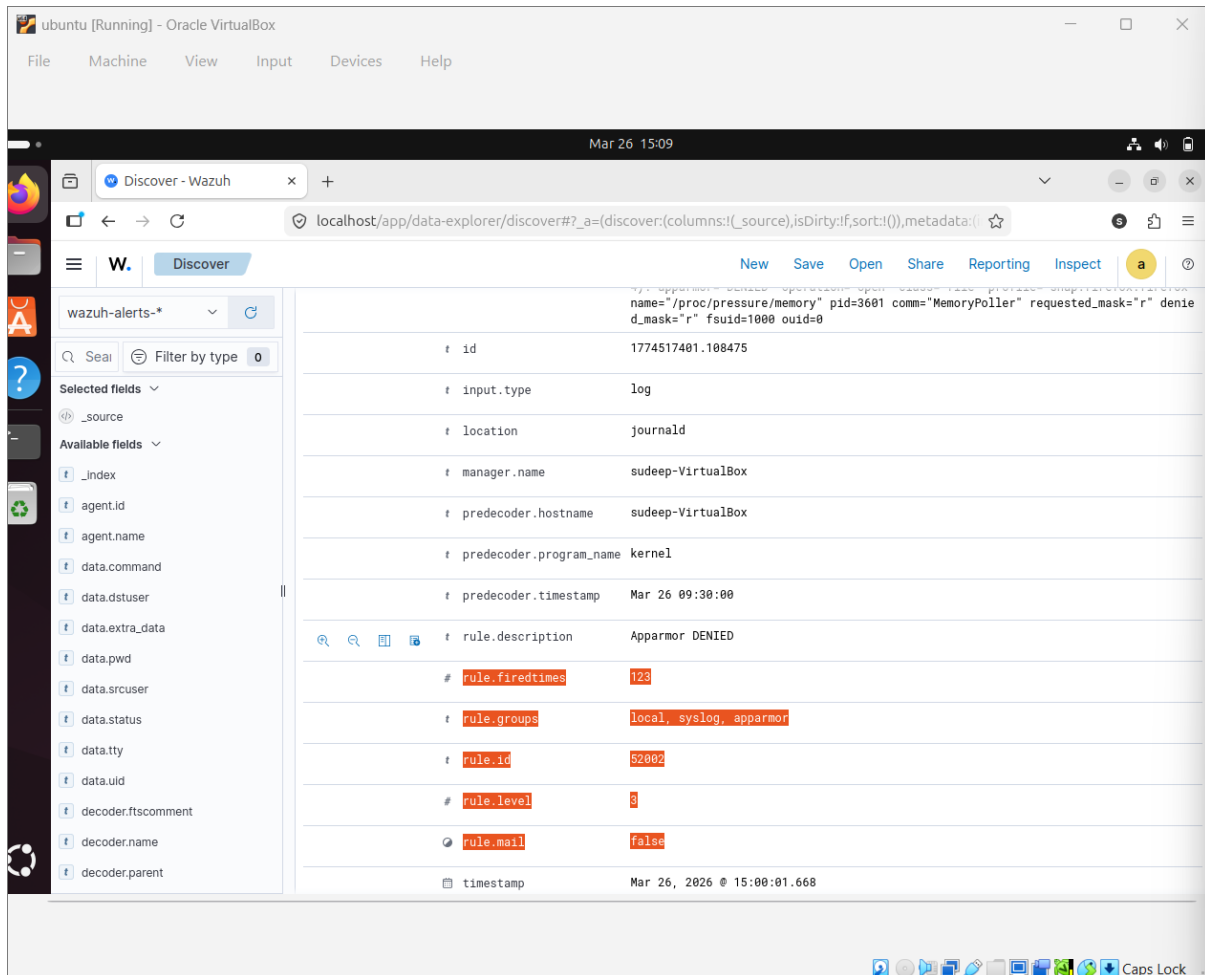


Figure 3 Wazuh Alert – Low Severity (AppArmor)-b

High Priority Alert

- Multiple failed SSH logins
- MITRE Mapping:
 - **T1110 – Brute Force**
- Indicates credential attack behavior

Classified as **Suspicious / Potential Attack**



The screenshot shows the Wazuh Discover interface in a web browser. The left sidebar contains a search bar and a list of available fields. The main panel displays a list of alerts, with the selected alert expanded to show its details in a table format.

Available fields:

- _index
- agent.id
- agent.name
- data.command
- data.dstuser
- data.extra_data
- data.pwd
- data.srcip
- data.srcuser
- data.status
- data.tty
- data.uid
- decoder.ftscment
- decoder.name

Alert Details Table:

Time	_source
Mar 26, 2026 @ 15:30:02.000	<pre>{ "predecoder.hostname": "sudeep-VirtualBox", "predecoder.program_name": "sshd", "predecoder.timestamp": "Mar 26 10:00:00", "input.type": "log", "agent.name": "sudeep-VirtualBox", "agent.id": "000", "manager.name": "sudeep-VirtualBox", "rule.mail": "false", "rule.level": 10, "rule.pci_dss": "10.2.4, 10.2.5", "rule.hipaa": "164.312.b", "rule.tsc": "CC6.1, CC6.8, CC7.2, CC7.3", "rule.description": "syslog: User missed the password more than one time", "rule.groups": "syslog, access_control, authentication_failed", "rule.nist_800_53": "AU.14, AC.7", "rule.gdpr": "IV.35.7.d, IV.32.2", "rule.firedtimes": 6 }</pre>

Expanded document Table:

Field	Value
_index	wazuh-alerts-4.x-2026.03.26
agent.id	000
agent.name	sudeep-VirtualBox
decoder.name	sshd
full_log	Mar 26 10:00:00 sudeep-VirtualBox sshd[19557]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
id	1774519202.207931
input.type	log
location	journald
manager.name	sudeep-VirtualBox

Figure 4 :Wazuh Alert – High Severity (SSH Brute Force)-a

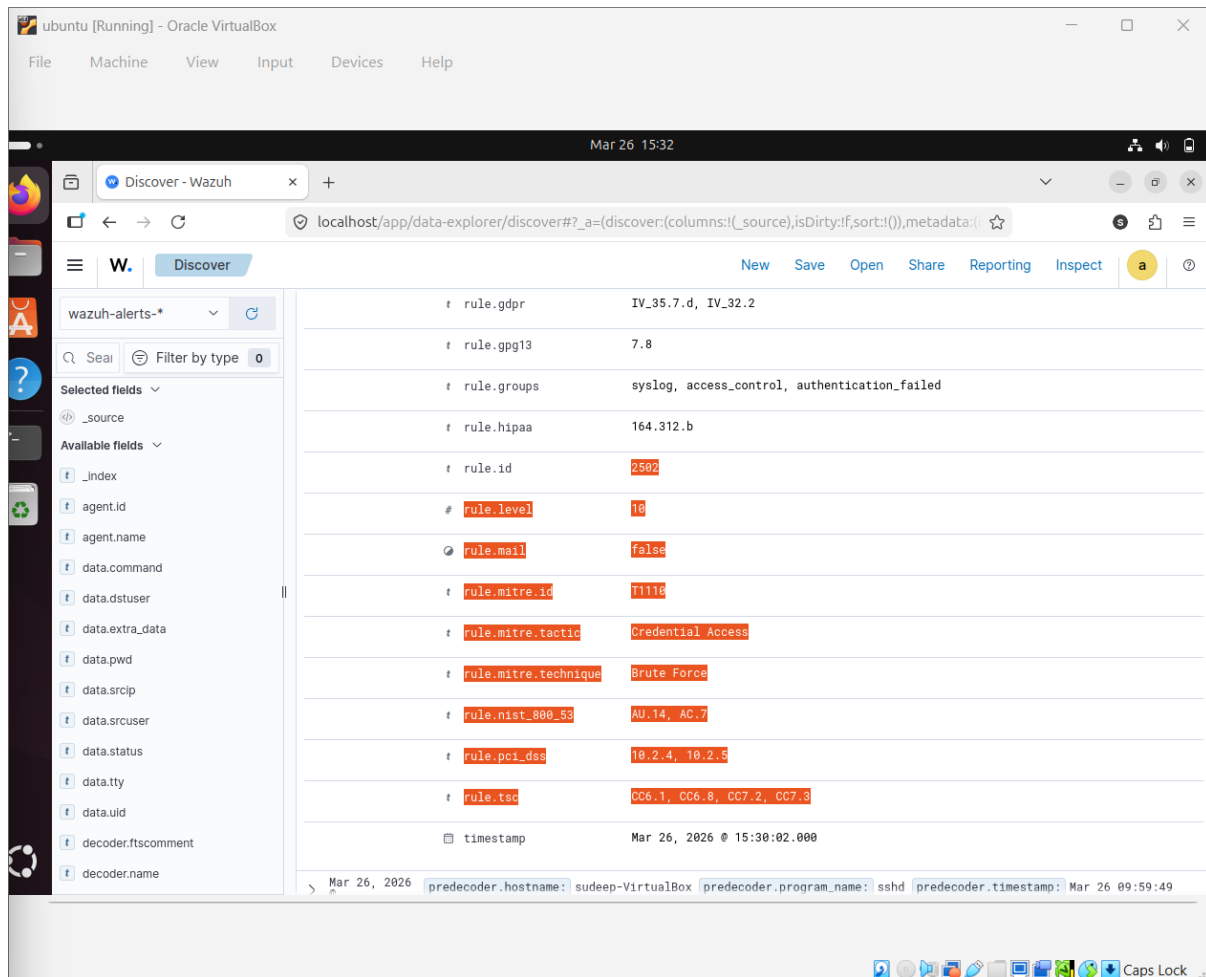


Figure 5 : Wazuh Alert – High Severity (SSH Brute Force)-b

Threat Intelligence Validation

VirusTotal Analysis

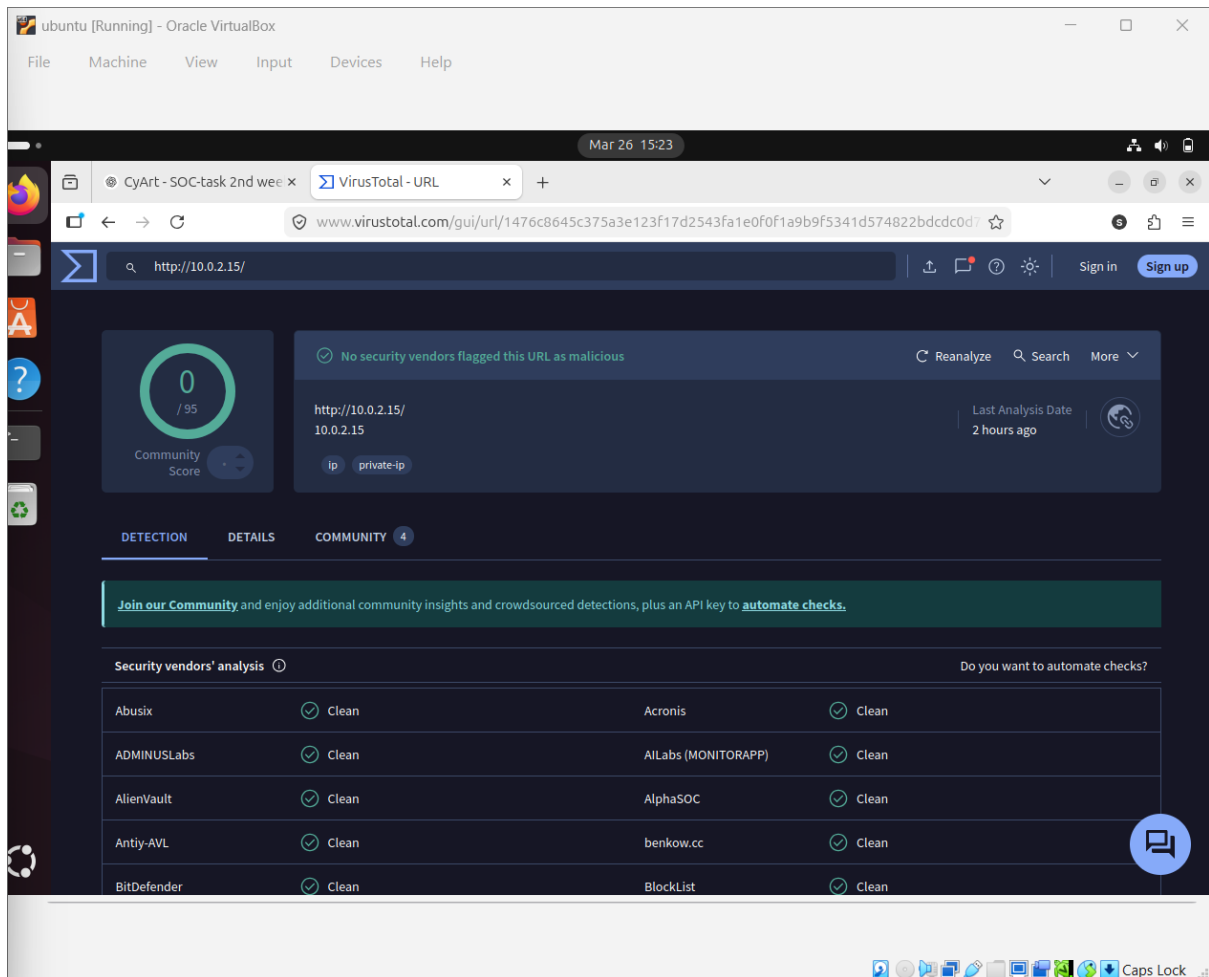


Figure 6 : VirusTotal Analysis – Local IP for low severity

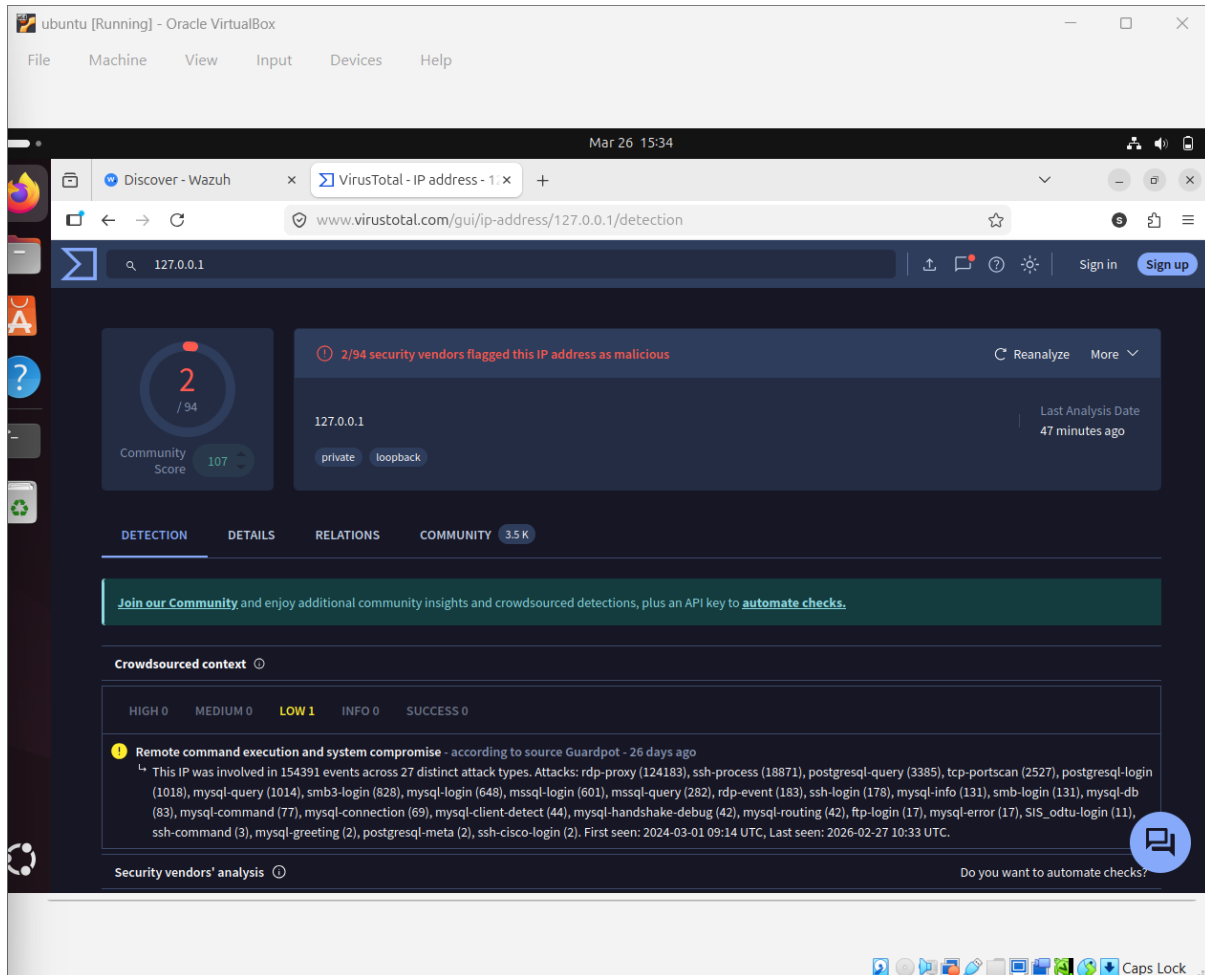


Figure 7 : VirusTotal Analysis – Local IP for High severity - a

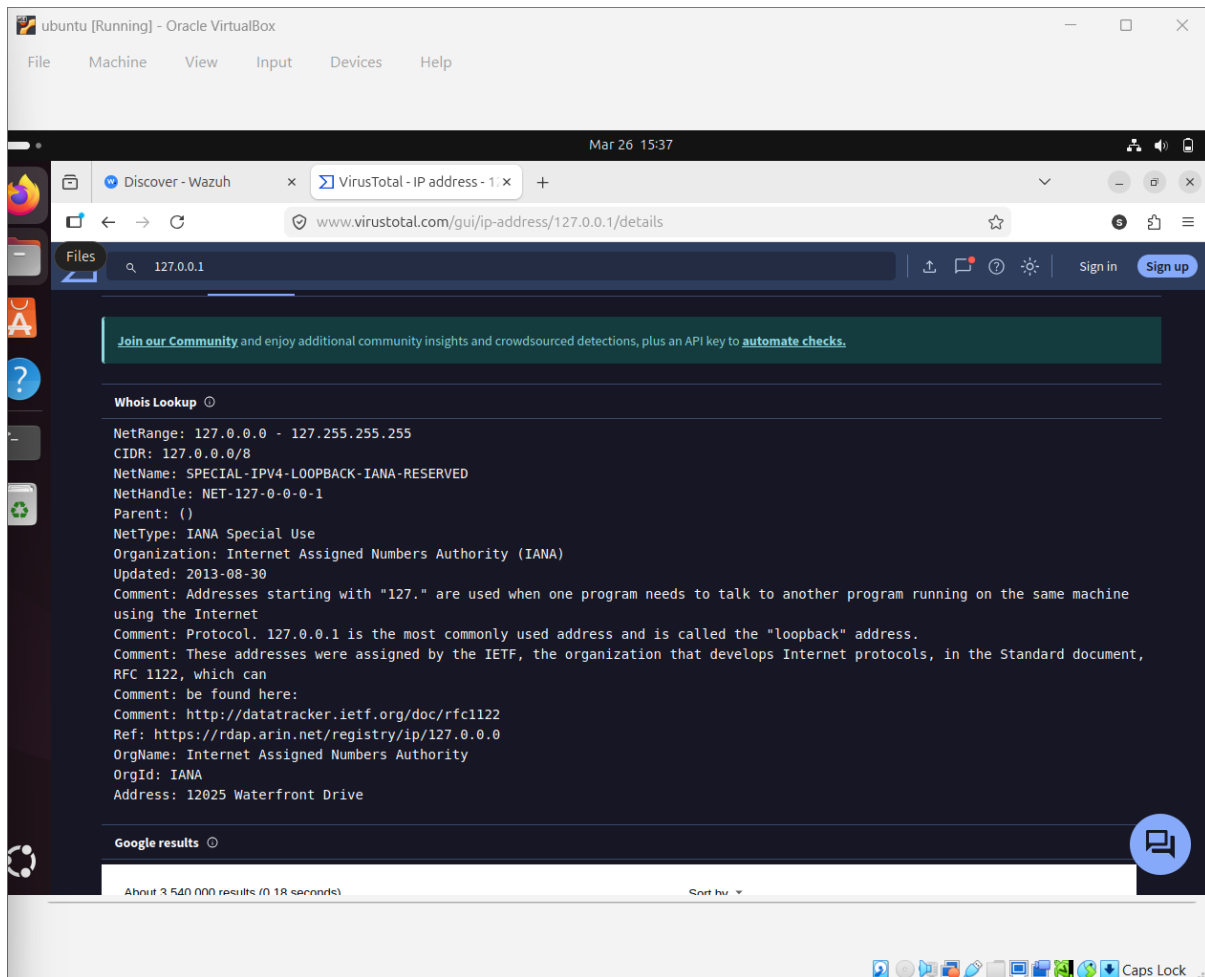


Figure 8 : VirusTotal Analysis – Local IP for High severity - b

Advanced Insight

Even though VirusTotal flagged some detections:

- It is due to **shared/global intelligence**
- Not relevant for localhost

Shows:

- Global attack patterns
- Attack types (SSH, RDP, SQL, etc.)
- Risk grading (A–E)

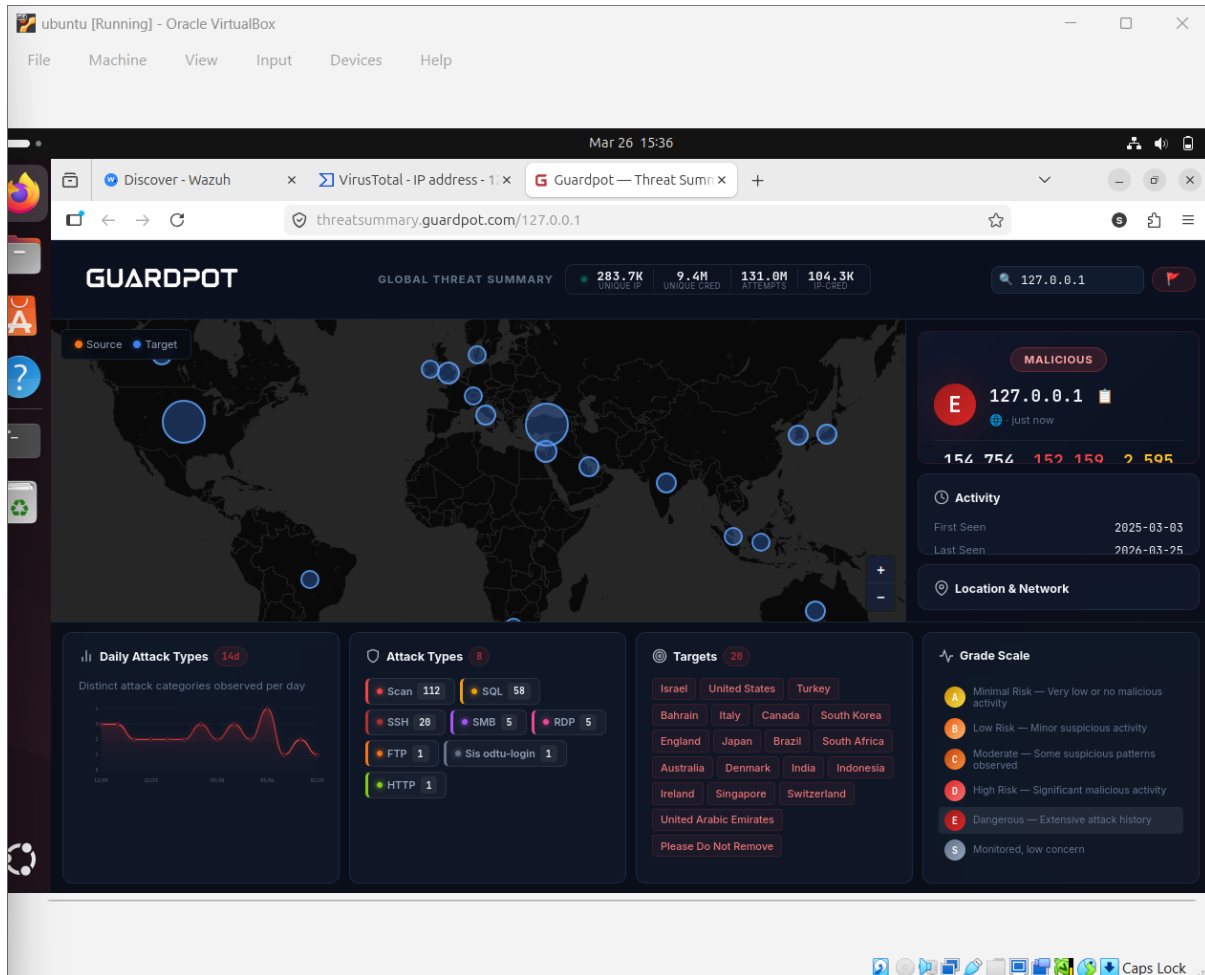


Figure 9 GuardPot Threat intelligence

“This exercise helped me understand how to distinguish between benign system activity and real attack patterns using SIEM correlation and threat intelligence validation.”

Conclusion

This task provided hands-on experience in alert triage and threat validation. It demonstrated how SOC analysts investigate alerts, assign priorities, and validate threats using intelligence platforms. The simulation of SSH brute-force attacks helped understand real-world attack detection and response workflow.



TASK 4: Evidence Preservation

Tools Used

- Velociraptor (for live data collection)
- Command Prompt (for hashing)
- Windows 10 Virtual Machine

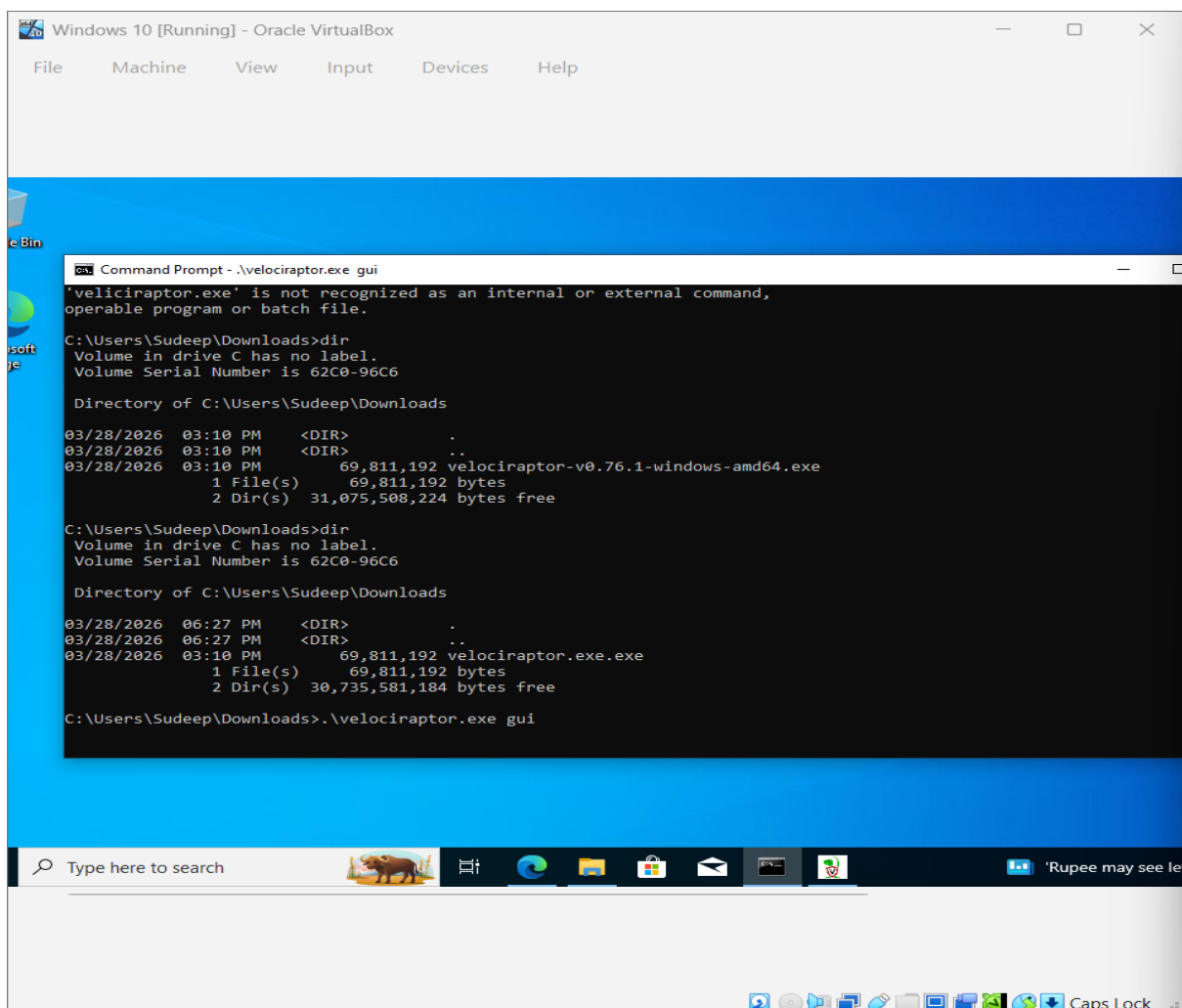
Part 1: Volatile Data Collection (Network Connections)

Volatile data such as active network connections was collected using Velociraptor with VQL query.

Start Velociraptor

- Open Command Prompt as Administrator
- Run:

velociraptor.exe gui



```
Windows 10 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

e Bin
Command Prompt - .\velociraptor.exe gui
'velociraptor.exe' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\Sudeep\Downloads>dir
Volume in drive C has no label.
Volume Serial Number is 62C0-96C6

Directory of C:\Users\Sudeep\Downloads

03/28/2026 03:10 PM <DIR> .
03/28/2026 03:10 PM <DIR> ..
03/28/2026 03:10 PM 69,811,192 velociraptor-v0.76.1-windows-amd64.exe
1 File(s) 69,811,192 bytes
2 Dir(s) 31,075,508,224 bytes free

C:\Users\Sudeep\Downloads>dir
Volume in drive C has no label.
Volume Serial Number is 62C0-96C6

Directory of C:\Users\Sudeep\Downloads

03/28/2026 06:27 PM <DIR> .
03/28/2026 06:27 PM <DIR> ..
03/28/2026 03:10 PM 69,811,192 velociraptor.exe.exe
1 File(s) 69,811,192 bytes
2 Dir(s) 30,735,581,184 bytes free

C:\Users\Sudeep\Downloads>.\velociraptor.exe gui
```

Figure 1 Velociraptor in CMD



Figure 2 : Server connection logs

Access Web Interface

- Open browser:

<https://127.0.0.1:8889>

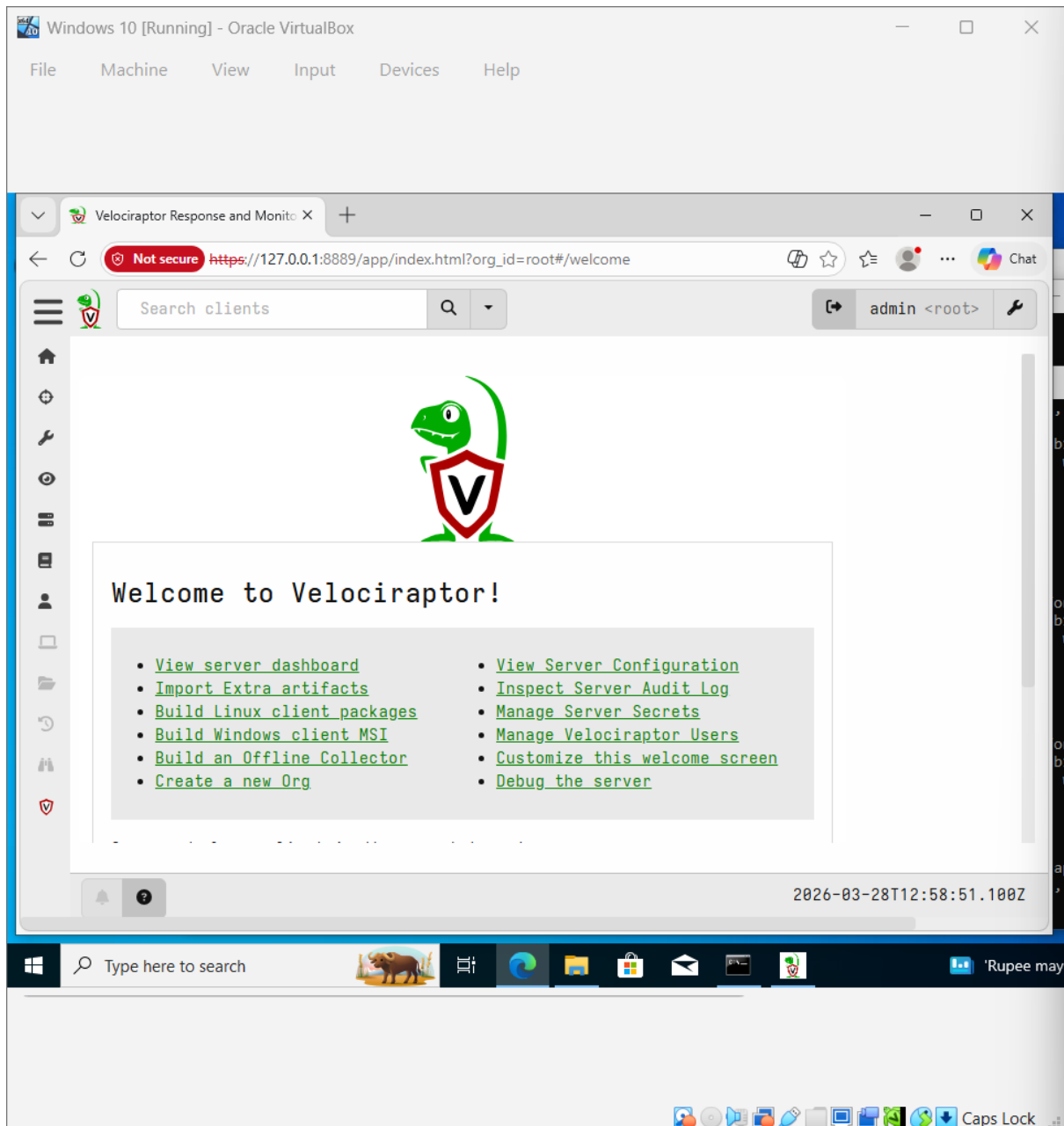


Figure 3 Velociraptor dashboard



Create Notebook

- Navigate to Notebook section
- Create new notebook:
 - Name: Netstat Investigation
 - Description: Collecting volatile network data

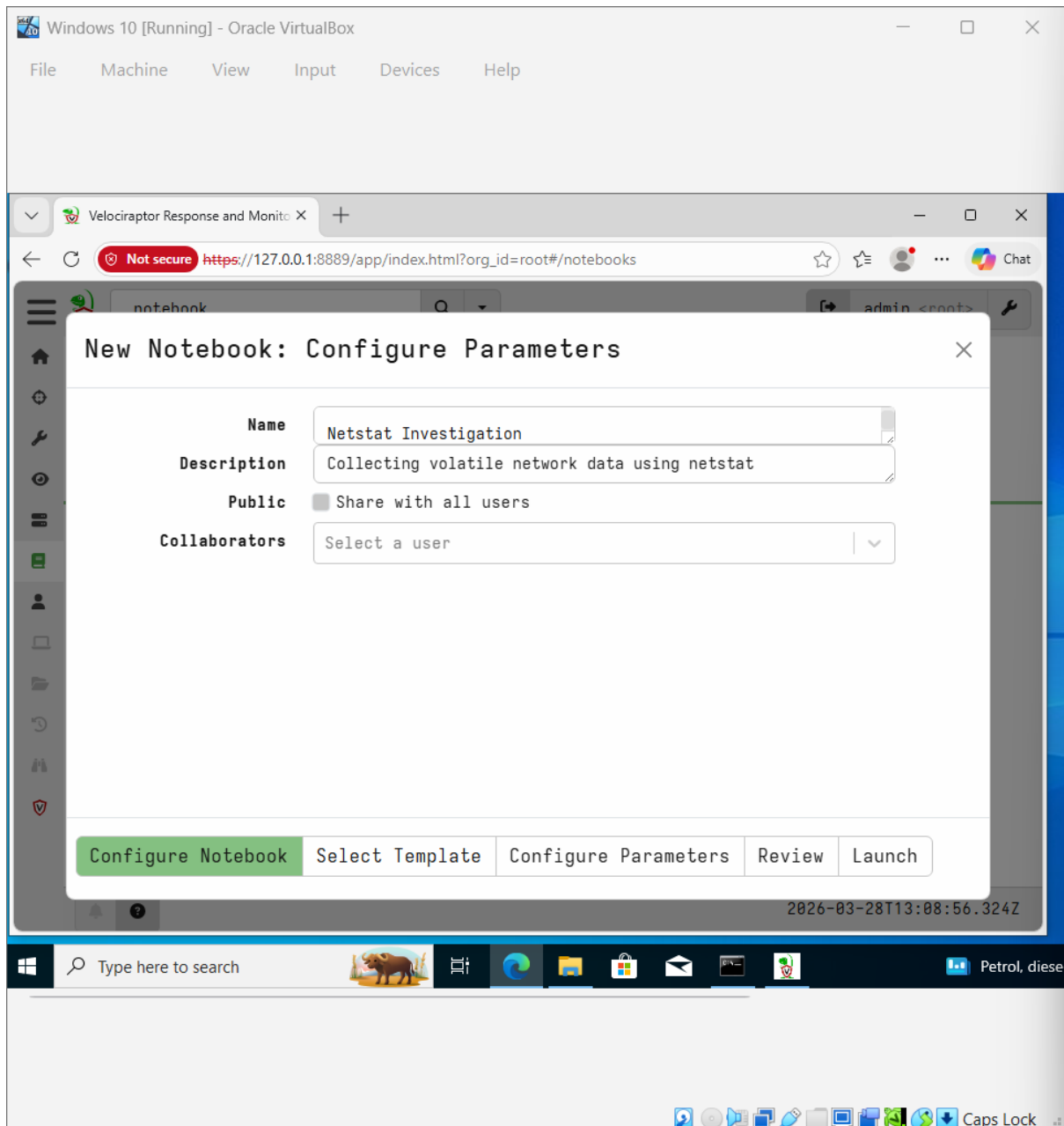


Figure 4: Notebook creation



Fd	Family	Type	Laddr	Raddr	Status	Pid	FamilyString	Timestamp
0	2	1	{ "IP": "0. 0.0.0" "Port": 1 35 }	{ "IP": "0. 0.0.0" "Port": 0 }	LISTEN	1012	IPv4	2026-03-28T12:49:56
0	2	1	{ "IP": "10 .0.2.15" "Port": 1 39 }	{ "IP": "0. 0.0.0" "Port": 0 }	LISTEN	4	IPv4	2026-03-28T12:49:56
0	2	1	{ "IP": "0. 0.0.0" "Port": 5 040 }	{ "IP": "0. 0.0.0" "Port": 0 }	LISTEN	3996	IPv4	2026-03-28T12:58:08
0	2	1	{ "IP": "12 7.0.0.1" "Port": 8 000 }	{ "IP": "0. 0.0.0" "Port": 0 }	LISTEN	6340	IPv4	2026-03-28T12:58:21

Figure 5 : Notebook netstat data -a

Fd	Family	Type	Laddr	Raddr	Status	Pid	FamilyString	Timestamp
0	2	1	{ "IP": "12 7.0.0.1" "Port": 8 000 }	{ "IP": "12 7.0.0.1" "Port": 4 9832 }	ESTAB	6340	IPv4	2026-03-28T12:58:21
0	2	1	{ "IP": "12 7.0.0.1" "Port": 8 000 }	{ "IP": "12 7.0.0.1" "Port": 4 9837 }	ESTAB	6340	IPv4	2026-03-28T12:58:21
0	2	1	{ "IP": "12 7.0.0.1" "Port": 8 001 }	{ "IP": "0. 0.0.0" "Port": 0 }	LISTEN	6340	IPv4	2026-03-28T12:58:21
0	2	1	{ "IP": "12 7.0.0.1" "Port": 8 001 }	{ "IP": "12 7.0.0.1" "Port": 4 9814 }	ESTAB	6340	IPv4	2026-03-28T12:58:21

Query Stats: {"RowsScanned":75,"PluginsCalled":1,"FunctionsCalled":1,"ProtocolSearch":0,"ScopeCopy":151}

Logs

Figure 6: Notebook netstat data -b



Figure 7.8.9 : CSV output file (query_1.csv)

Result

- Active network connections were successfully collected
- Data exported as **CSV file (query_1.csv)**
- Contains:
 - IP Address
 - Port
 - Status (LISTEN / ESTABLISHED)
 - Process ID



Part 2: Memory Evidence Collection

Memory dump was collected to capture running processes and system state.

Select Artifact

- Navigate to **Collections**
- Select:

Windows.Memory.ProcessDump

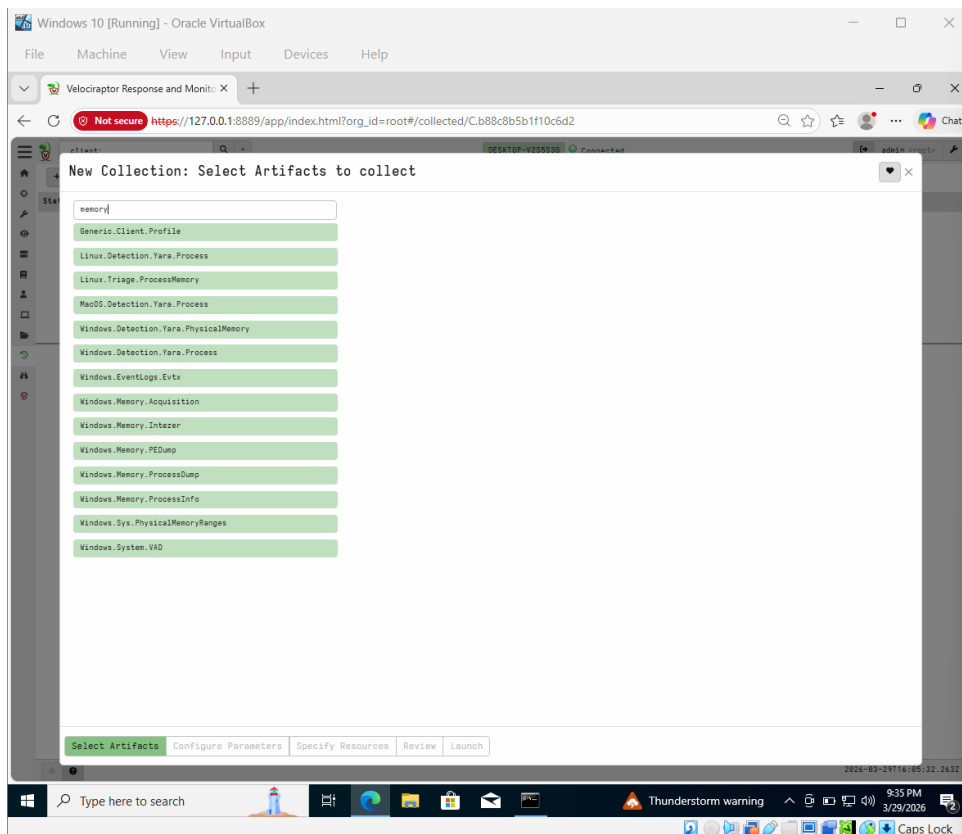


Figure 10 : Selecting memory mini dump

2. Execute Collection

- Run artifact on client system

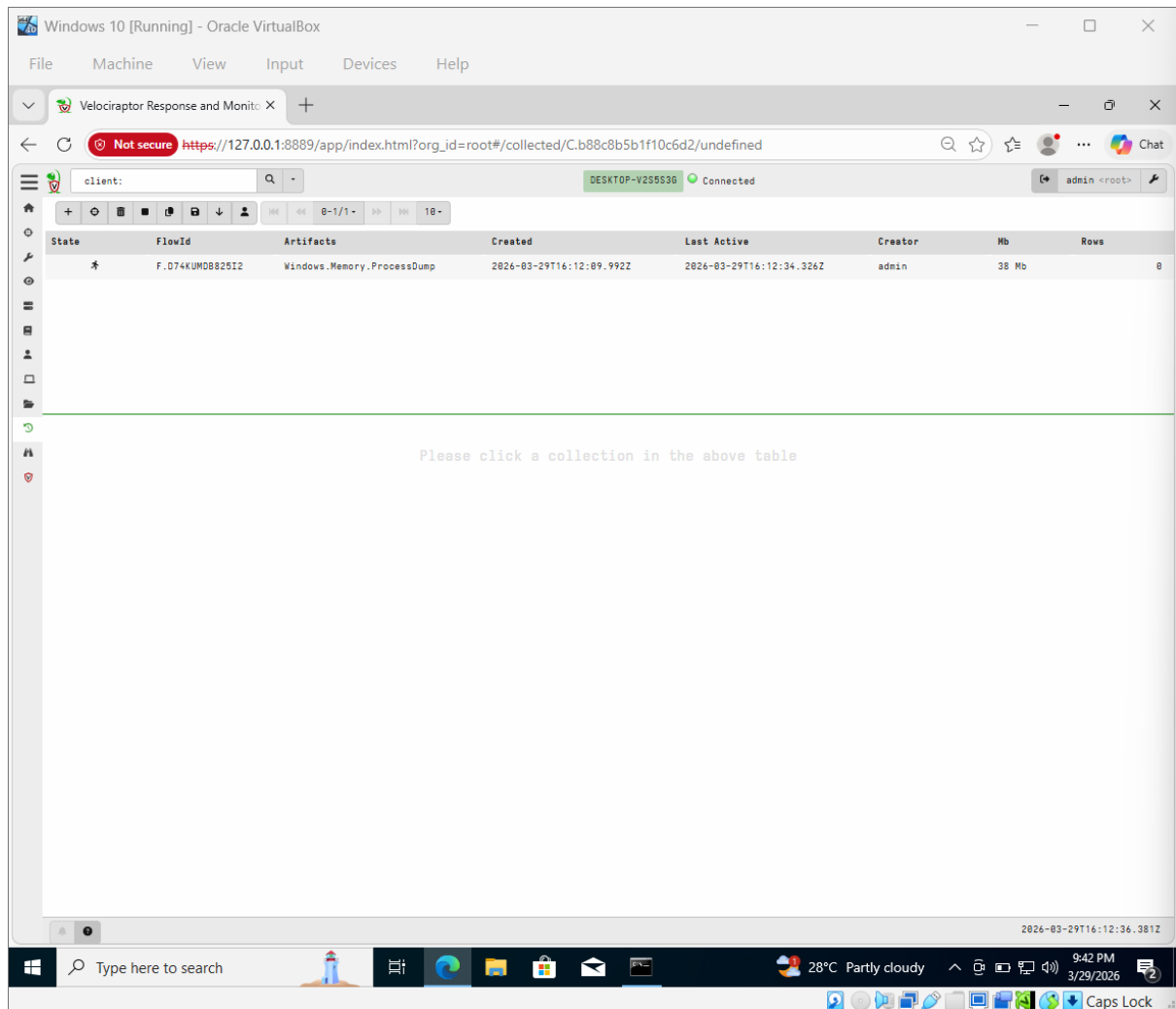


Figure 11 : Host information



The screenshot shows the Velociraptor Response and Monitor web interface running in a Windows 10 virtual machine. The browser address bar shows a URL: https://127.0.0.1:8889/app/index.html?org_id=root#/collected/C.b88c8b5b1f10c6d2/F.D74KUMDB882512/uploads. The interface displays a table of collected artifacts and a detailed view of uploaded files.

State	FlowId	Artifacts	Created	Last Active	Creator	Mb	Rows
*	F.D74KUMDB882512	Windows.Memory.ProcessDump	2026-03-29T16:12:09.992Z	2026-03-29T16:14:03.172Z	admin	38 Mb	8

Timestamp	started	vfs_path	Type	file_size	uploaded_size	Preview
2026-03-29T16:12:11Z	2026-03-29 16:12:11.8699587 +0800 UTC	4_System.dmp		0	0	
2026-03-29T16:12:11Z	2026-03-29 16:12:11.8888099 +0800 UTC	132_Registry.dmp		0	0	
2026-03-29T16:12:11Z	2026-03-29 16:12:11.1321923 +0800 UTC	416_smss.exe.dmp		0	0	
2026-03-29T16:12:11Z	2026-03-29 16:12:11.1583217 +0800 UTC	520_csrss.exe.dmp		0	0	
2026-03-29T16:12:11Z	2026-03-29 16:12:11.1807336 +0800 UTC	596_csrss.exe.dmp		0	0	
2026-03-29T16:12:11Z	2026-03-29 16:12:11.2031502 +0800 UTC	604_wininit.exe.dmp		0	0	
2026-03-29T16:12:14Z	2026-03-29 16:12:14.2296902 +0800 UTC	668_winlogon.exe.dmp		39968033	39968033	MDMP 5a ...
2026-03-29T16:12:14Z	2026-03-29 16:12:14.2475258 +0800 UTC	740_services.exe.dmp		0	0	

Figure 12 : Memory Dump files generated

Part 3: Hashing

To ensure integrity of evidence, SHA256 hashing was performed.

Navigate to file location

```
cd %USERPROFILE%\Downloads
```

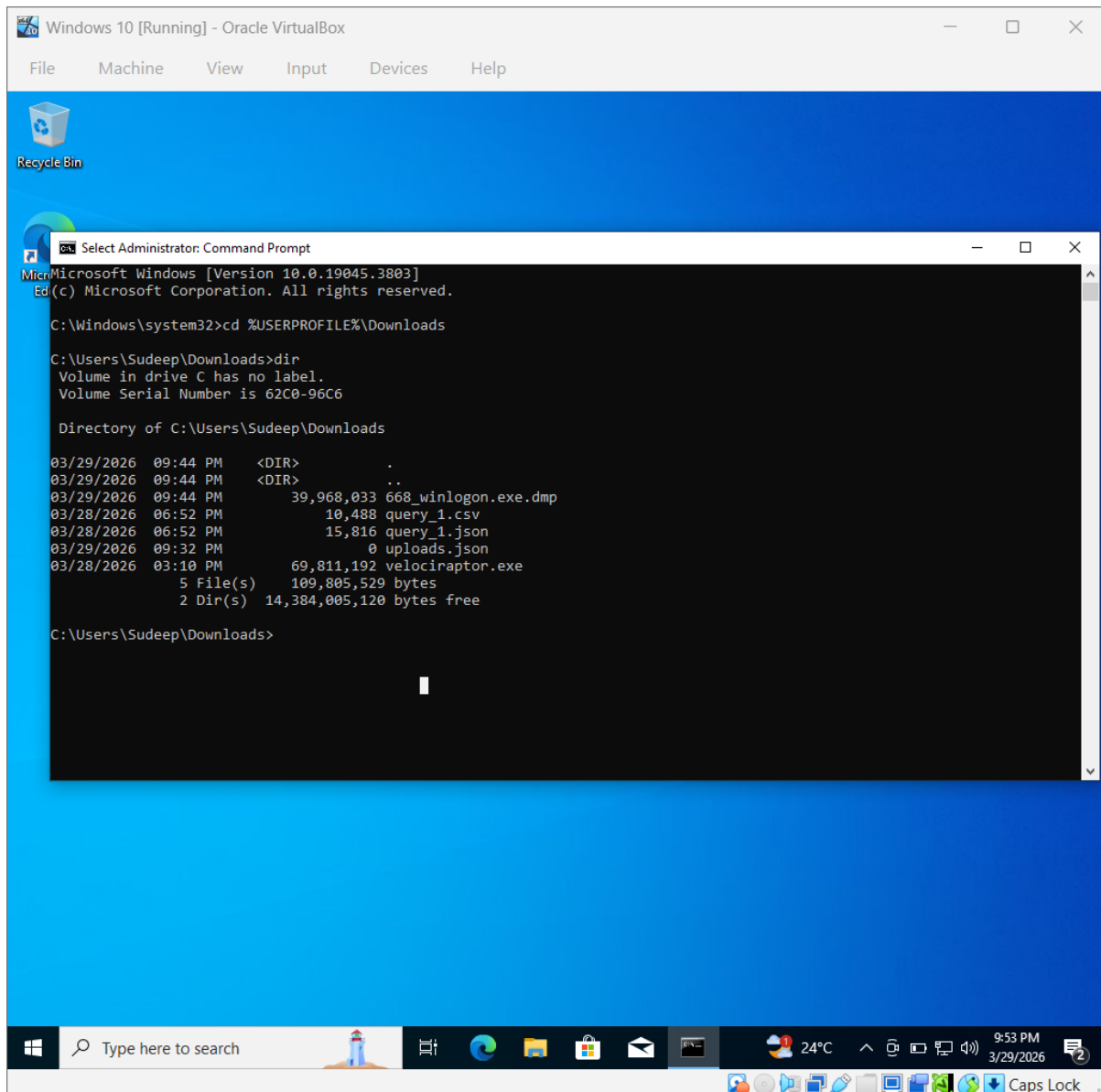


Figure 13 : Downloaded file

Generate Hash

certutil -hashfile 668_winlogon.exe.dmp SHA256



Figure 14: Hash generated output

Result

SHA256 Hash:

8163078ced03ae846f0c6a39c497ac1b2d65a148b2d8888ed1668ccaf0e51357



Evidence Preservation Table

Item	Description	Collected By	Date	Hash Value
Memo	winlogon.exe	SOC	2026-	8163078ced03ae846f0c6a39c497ac1b2d65a148b2d8888ed
ry	Process	Analyst	03-	1668ccaf0e51357
Dump	Dump		29	

Conclusion

- Volatile network data was successfully collected using Velociraptor
- Memory dump evidence was acquired from the system
- Evidence integrity was verified using SHA256 hashing
- Proper documentation was maintained for chain-of-custody

TASK 5. Capstone Project: Full Alert-to-Response Cycle